SVD-Compressed Cross Sections in Monte Carlo Neutron Transport

An implementation-led benchmark with a partition-of-unity fix for off-library temperature interpolation

F. Rodrigues
fabio.rodrigues@outsystems.com

May 4, 2026

Abstract

A pure-Rust Monte Carlo neutron transport engine implements four cross-section providers behind a common interface — pointwise table, truncated SVD of $\log_{10}\sigma(E,T)$, ACE+WMP, and SVD+WMP hybrid — and is used to characterise SVD compression as an alternative to pointwise tables, with a partition-of-unity variant of Ducru's kernel reconstruction [1] for off-library operating temperatures. On the Godiva HEU-MET-FAST-001 fast critical benchmark [2] the engine's rank-5 SVD provider reproduces the measured $k_{\text{eff}} = 1.0000 \pm 100$ pcm (1σ) to $\Delta_{\text{exp}} = +79 \pm 38$ pcm, inside the experimental uncertainty band; OpenMC 0.15.3 on the same ENDF/B-VII.1 HDF5 files sits at −99 pcm, also inside the band, with the two codes straddling experiment from opposite sides. On a 9-nuclide PWR pin cell (no physical benchmark exists; OpenMC 0.15.3 is used as an independent code-to-code reference) all four in-engine providers agree with OpenMC to $|\Delta k| \le 51$ pcm at 10-seed × 150-batch × 50 000-particle statistics. A single-seed four-way comparison on the same problem (Table / SVD / Hybrid SVD+WMP / ACE+WMP) puts every gap below $1\sigma_{\text{seed}}$: SVD-vs-WMP −4 pcm, Hybrid-vs-WMP −26 pcm, Table-vs-WMP +82 pcm, all at $\sigma_{\text{seed}} \approx 100$ pcm. 43 of 47 non-redundant ^{235}U reaction channels are rank-one to machine precision, and rank-5 SVD with discrete-level rank-1 recovers engine-wide k_{eff} within ∼ 50 pcm of the in-engine pointwise baseline on both benchmarks at 1.37×–1.90× the CPU throughput. At a realistic off-library operating point (+150 K above every nuclide's library column, fuel 900 → 1050 K, moderator 600 → 750 K), a partition-of-unity 3-point Ducru scheme reproduces k_∞ within 213 pcm (∼ 3.6σ) of the ACE+WMP industry baseline while using 22% less memory and matching its transport speed — a 10× reduction over a naive 2-point bracketing variant that leaves a 2094 pcm kernel-approximation bias.

The hybrid memory picture in the engine is now measured end to end: a smooth-only SVD rebuild outside each WMP window (MT= 2, 18, 102) drops the live working set from 519.0 MB to 487.6 MB, a 1.06× reduction. The 132.9× representation-byte ratio against a four-channel pointwise table reported in the literature is correct on its own terms but is not the engine-scale reduction; this paper documents both numbers and explains why they differ.

Three limitations are reported quantitatively: GPU SVD is 1.3× slower than GPU pointwise on PWR because four clad nuclides require summing 13 discrete-level SVD kernels per lookup; the WMP+SVD hybrid is 2× slower than CPU SVD because Faddeeva evaluation is ∼ 15× more expensive than the SVD FMA; and a simplex-projected QP variant of Ducru (clip negative weights, renormalize) degrades U-238 capture reconstruction from 0.84% to 5% L2 error because the projection is optimal in weight space rather than in the Doppler-kernel norm. The kernel-Gram QP from [1] §4 wins at rank 3 (0.60% L2) but loses at production rank 5 (2.06%); signed three-point unity is the default at rank 5.

Keywords: Monte Carlo, cross-section compression, singular value decomposition, windowed multipole, Doppler broadening, partition-of-unity interpolation, nuclear data, GPU transport.

1 Introduction

Continuous-energy Monte Carlo codes such as OpenMC [3], MCNP, and Serpent store nuclear data as pointwise tables. Per reaction channel and per library temperature, a piecewise-linear or log-log interpolation array carries 10^4–10^5 grid points for light nuclides and 10^5–10^6 for dense-resonance actinides. A full-library actinide set at a handful of temperatures exceeds the working-set of contemporary L2/L3 caches, and every collision pays for a binary search plus a log-log blend through a large monotonic array.

Prior work attacks this bottleneck along three axes. Adaptive resampling machine-learned models retain 10–50% of the pointwise data at ≤ 97 pcm criticality error [4, 5]. The Windowed Multipole (WMP) method [6–8] gives an analytical pole–residue representation of the resolved resonance region with on-the-fly Doppler broadening. GPU rewrites reorganise the transport loop to amortise the lookup across warps [9, 10].

This paper asks a narrower question. Can a truncated SVD of the $\log_{10}$-transformed matrix $\mathbf{A} \in \mathbb{R}^{N_E \times N_T}$ replace the pointwise table as a drop-in cross-section representation, and if so, at what cost? We implemented four providers (pointwise table, SVD kernel, ACE+WMP hybrid, and SVD+WMP hybrid) inside a single pure-Rust transport engine, plus matching CUDA paths for pointwise and SVD (and a standalone CUDA WMP evaluator), and measured fidelity, throughput, and memory side-by-side.

The answer, reported in full in Sections 6 and 10, is *qualified*: SVD is a defensible drop-in on the thermal pin cell tested here; the hybrid implementation works end-to-end and agrees with the ACE+WMP industry baseline to −26 pcm at single-seed statistics (well inside seed noise); the throughput picture is hardware-dependent and favours pointwise tables on our GPU; and the memory picture at engine scale is *not* the 132.9× reduction a representation-byte analysis suggests — in the live engine the smooth-only SVD rebuild that excludes WMP-covered energies trims the working set from 519.0 MB to 487.6 MB, a 1.06× reduction. We document both numbers and explain why they differ.

Contributions.

- A singular-spectrum measurement showing 43 of 47 non-redundant ^{235}U reactions are rank-one to machine precision at $T = 294$ K across six library temperatures, with a per-reaction distribution and Pareto (Sec. 4, Fig. 1).

- A rank-k SVD reconstruction kernel for pointwise cross sections, with Ducru temperature interpolation [1]. Rank-6 reaches machine precision at 44 ns/lookup against 91 ns for a binary-search pointwise table (kernel-level 2.0×).

- A PWR pin cell rank sweep (9 nuclides, S(α,β), URR, discrete levels) showing CPU SVD rank-5 matches the in-engine pointwise table on k_∞ to 13–47 pcm on two hardware tiers at 1.37×–1.90× throughput (Sec. 6).

- A matching event-based CUDA solver [9] for both providers, showing GPU pointwise is 1.30× faster than GPU SVD on PWR pin cell, inverting the CPU ordering (Sec. 9).

- A hybrid SVD+WMP engine: pure-Rust Faddeeva reader and evaluator matching OpenMC Python to $\sim 10^{-6}$ relative error, wired into a `HybridSvdWmpXsProvider`. On the PWR pin cell at single-seed statistics, $k_\infty = 1.32795 \pm 0.00103$ (-26 pcm vs ACE+WMP, $\ll 1\sigma_{\text{seed}}$). Hybrid throughput is $\sim 2\times$ slower than CPU SVD on the large-L3 reference run; in-engine smooth-only memory 1.06× smaller than the full-SVD baseline (Sec. 10).

- A CUDA port of the WMP evaluator, bit-exact against the CPU path ($\leq 5 \cdot 10^{-14}$ relative at double precision), with 7.0× GPU-vs-CPU per-lookup throughput on an RTX 3080 (Sec. 11).

- Ten-seed PWR k_∞ on all four in-engine providers agrees with OpenMC 0.15.3 to $|\Delta| \leq 51$ pcm (CPU table -12, CPU SVD -25, GPU pointwise $+51$, GPU SVD -8). The angular and fission outgoing-energy distributions use stochastic-bin sampling on both CPU and GPU; a Shannon-entropy source-convergence monitor is provided as a runtime diagnostic (Sec. 6.4).

- Validation against the ICSBEP Godiva HEU-MET-FAST-001 *physical* benchmark $k_{\text{eff}}^{\text{exp}} = 1.0000 \pm 100$ pcm (1σ) [2]. The rank-5 SVD provider reproduces the experimental value to $\Delta_{\text{exp}} = +79 \pm 38$ pcm, inside the experimental uncertainty band. OpenMC 0.15.3 on the same ENDF/B-VII.1 HDF5 files sits at -99 pcm, also inside the band; the two codes straddle the measurement from opposite sides (Sec. 5.1).

- A clear separation of two memory measurements: the WMP representation-byte ratio against a hypothetical four-channel pointwise table is 132.9× for our nine-nuclide set, and the live in-engine reduction from smooth-only-rebuild hybrid over full-SVD is 1.06× (519.0 → 487.6 MB, measured on the loaded kernels at run time). The hybrid engine in absolute terms is 4.8–5.1× larger than the pointwise-table baseline, because discrete-level SVD kernels dominate memory and are not covered by the public WMP library. The gap between representation-byte ratios and engine-scale savings is the main cautionary finding of this paper, and we discuss what would need to be true for a WMP-scale reduction to land at engine level (Sec. 10.1, Sec. 10.4).

2 Method

2.1 Pointwise baseline

For each nuclide and library temperature, $\sigma(E_i, T_j)$ lies on a union energy grid $\{E_i\}_{i=0}^{N_E-1}$. For $E_i \leq E < E_{i+1}$ the standard log-log lookup is

$$\sigma(E) = \sigma_i \,(\sigma_{i+1}/\sigma_i)^f, \qquad f = \frac{\ln(E/E_i)}{\ln(E_{i+1}/E_i)}. \tag{1}$$

We use a log-uniform hash index with 8192 bins (Brown 2014 [11]) to resolve i in $O(1)$, falling back to a bounded binary search in dense bins. Non-positive cross-section values fall back to linear interpolation to avoid log of a negative.

2.2 Truncated SVD

Pre-clamp the matrix to avoid $-\infty$ in the logarithm, $\mathbf{A}_{ij} = \max(\sigma(E_i, T_j), 10^{-30})$, and compute the thin SVD of $\mathbf{L} = \log_{10}\mathbf{A}$:

$$\mathbf{L} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top, \qquad \mathbf{L} \approx \mathbf{U}_k\boldsymbol{\Sigma}_k\mathbf{V}_k^\top. \tag{2}$$

Pre-multiply the basis $\mathbf{B} = \mathbf{U}_k\boldsymbol{\Sigma}_k$ so a reconstruction at grid point i for target coefficient vector $\mathbf{c} \in \mathbb{R}^k$ is one dot product:

$$\log_{10}\tilde{\sigma}(E_i, T) = \sum_{\ell=1}^{k} B_{i\ell}\, c_\ell, \qquad \tilde{\sigma} = 2^{(\log_{10}\tilde{\sigma})\cdot\log_2 10}. \tag{3}$$

$\mathbf{B}$ stores $N_E k$ doubles row-major so k values for one energy point live in $8k$ contiguous bytes; at $k = 5$ that is one cache line. Between grid points we apply log-log interpolation on $\log_{10}\tilde{\sigma}$ and take one $\exp_2$.

2.3 Temperature interpolation

Library temperatures are typically $\{0, 250, 294, 600, 900, 1200, 2500\}$ K in the ENDF/B-VII.1 HDF5 distribution. For target $T \notin \{T_j\}$ the coefficient vector follows the Ducru kernel method [1]:

$$c_\ell(T) = \sum_{j\in\mathcal{S}} w_j(T)\, V_{j\ell}, \qquad w_j(T) = \frac{\sqrt{T_j T}}{T_j + T} \prod_{i\in\mathcal{S}, i\neq j} \frac{T - T_i}{T + T_i}\cdot\frac{T_j + T_i}{T_j - T_i}, \tag{4}$$

where $\mathcal{S}$ is a subset of library temperatures used for the interpolation. Eq. 4 is exact at library temperatures (w collapses to a one-hot selector when $T = T_j$), $\sim 0.1\%$ error on Doppler-broadened cross sections over 300–3000 K, and L2-optimal for the free Doppler kernel on $\mathcal{S}$.

Choosing the interpolation subset

Using $|\mathcal{S}| = N$ (all library temperatures) is numerically unstable for $N \gtrsim 4$ because the product-of-ratios term in

w_j develops near-singular factors when non-adjacent T_i's bracket T with large stride. In our measurements with $N = 7$ the 2500 K column pulls the product toward values $\sim 10^2$, the resulting weights routinely swing by a factor of 10–100 across neighbouring target temperatures, and the log-space reconstruction sees non-physical oscillations that scale linearly with the condition number.

We use the three library temperatures nearest to T. Three is the smallest subset that captures a quadratic correction to the Faddeeva kernel (two-point misses it; see Sec. 8); it is also small enough that the product term has only two factors per weight and the formula is well-conditioned at every operating temperature we tested in the interval $[250, 2500]$ K. Section 2.4 describes how we make the subset weights a partition of unity; this is essential for resonance-dominated systems and is the change that brings the off-library PWR result into the deployable band against ACE+WMP.

2.4 Partition-of-unity Ducru for off-library T

When Eq. 4 is applied to $V^\top$ columns of a $\log_{10}\sigma$ SVD, the reconstructed $\log_{10}\tilde{\sigma}(E_i, T)$ is a linear combination of the per-library-T log cross sections with coefficients $w_j(T)$. On the *linear* cross section this is a weighted geometric mean:

$$\tilde{\sigma}(E, T) = \prod_{j\in S} \sigma(E, T_j)^{w_j(T)}. \tag{5}$$

If $\sum_j w_j \neq 1$, the right-hand side of Eq. 5 is scaled uniformly by $\exp[(\sum_j w_j - 1)\cdot\overline{\log\sigma}]$ where the overbar denotes a weighted average of $\log_{10}\sigma(E, T_j)$. For smooth channels this is a barely-visible gain error (far away from resonance peaks, the average $\log\sigma$ is small and the multiplicative shift is close to unity). On narrow resonance peaks it is catastrophic: at the U-238 6.67 eV resonance with a peak cross section of ~ 4500 b, a 5% gain error moves the peak by ~ 225 b, which on a PWR pin cell translates to ~ 2000 pcm on k_∞ (Sec. 8).

Raw Ducru does *not* have $\sum_j w_j = 1$ in general. Eq. 4's weights are designed to reconstruct $\sigma(E, T)$ in the free-Doppler-kernel L2 norm (see [1], §3); partition of unity is not one of the optimization's constraints. Measured on our $N = 3$ bracketing subset at $T = 750$ K with library $(600, 900, 1200)$ K the raw weights $(0.21, 0.30, -0.01)$ sum to 0.50. At $T = 1050$ K with library $(900, 1200, 2500)$ K they sum to 0.51. At endpoints they collapse to $(1, 0, 0)$ / $(0, 1, 0)$ so the total is 1; *between* endpoints the sum drifts by up to ≈ 0.5.

We restore partition of unity by a single division:

$$\tilde{w}_j(T) = \frac{w_j(T)}{\sum_{k\in S} w_k(T)}, \qquad \sum_{j\in S} \tilde{w}_j(T) = 1. \tag{6}$$

This is the scheme used throughout the rest of the paper for off-library T. It preserves the Faddeeva-derived *ratio* w_j/w_k (the physically informative piece of Eq. 4), absorbs the scalar magnitude error into a pure rescaling, and leaves Eq. 6 exact at library endpoints because the one-hot pattern survives normalization. Sec. 8 measures the k_∞ consequence on PWR and Godiva.

What $\tilde{w}_j$ is *not*

It is not the Ducru QP from [1] §4. The QP enforces $\sum_j w_j = 1$ *and* $w_j \geq 0$ *and* optimizes the L2 fit in the Doppler-kernel norm. Eq. 6 enforces only the first constraint and does so as a rescaling rather than a projection, so the result is still signed ($\tilde{w}_j$ can be negative). On U-238 MT= 102 held out at 900 K, the non-negative simplex projection [12] of the raw Ducru weights raises the L2 reconstruction error from 1.5% to 5% relative to the signed scheme (Sec. 8.4). For the benchmarks reported here, Eq. 6 suffices; an analytic kernel-QP is left to future work.

2.5 Memory trade-off of truncated SVD

Truncated SVD is not a free memory win, and the comparison against a pointwise table depends on how many temperatures the library stores. A single-temperature pointwise table stores one value per energy point per reaction. An SVD basis at rank k stores k values per energy point per reaction regardless of the number of library temperatures it was trained on. Writing T for the stored temperature count:

$$\frac{\text{SVD bytes}}{\text{pointwise bytes}} = \frac{N_E \cdot k}{N_E \cdot T} = \frac{k}{T}. \tag{7}$$

So:

- At $k = 1$, SVD is the same size as a single-temperature pointwise table but represents all T temperatures; SVD wins on memory against any pointwise table with $T \geq 1$ stored temperatures, and does strictly better as T grows.
- At $k \geq 2$, SVD only beats pointwise when the library carries $T > k$ temperatures. Break-even sits around $T = 2$–3. For a single-temperature toy problem SVD is strictly larger.
- Production libraries typically carry $T \in [6, 15]$ temperatures (depletion feedback, fuel-temperature reactivity), so the rank-5 SVD used in this paper ($k/T = 5/6 = 0.83$ to $5/15 = 0.33$) is a modest to strong memory win in the regime that matters, and a loss in the regime that does not.

The cost side of that trade-off is not zero:

- **Compute per lookup.** Pointwise is a hashed index plus one log-log interpolation. SVD is a rank-k FMA sequence plus a Ducru temperature step (Sec. 2.3). At $k = 5$ on the large-L3 system, SVD is 43 ns/lookup against 91 ns for the table (Sec. 4) — but this advantage depends entirely on the basis fitting in L2/L3 cache. A higher-rank basis or a larger nuclide set could tip this back toward pointwise.
- **Reconstruction error.** Pointwise is exact at grid points and piecewise-linear-in-log between them; SVD is an approximation everywhere. RMSE on $\log_{10}\sigma$ is $6\cdot 10^{-4}$ at $k = 5$ and reaches machine precision at $k = 6$ (Sec. 4). Lower rank means smaller memory and faster lookup at the cost of a tunable amount of cross-section error, which propagates into k_{eff} as pcm-scale bias.

- **One-time decomposition cost.** SVD requires an offline factorisation per nuclide per reaction (seconds at load time). Pointwise is a direct HDF5 read. Not a practical issue but not free either.
- **Temperature-interpolation quality.** Both representations interpolate between library temperatures at runtime. Neither performs on-the-fly Doppler broadening from first principles — WMP is the only method of the three in this paper that does. SVD with Ducru kernels gives $\sim 0.1\%$ error on broadened cross sections over 300–3000 K, which is tight but not analytical.
- **Rank dispersion across reactions.** Most reactions are rank-one to machine precision (43 of 47 non-redundant ^{235}U channels, Sec. 4.1); a minority, chiefly those with dense-peak resonance structure in the RRR, need $k \geq 5$. Using a uniform rank across all reactions wastes bytes on the easy channels; per-reaction rank would recover those but complicates the dispatch.
- **No safe extrapolation outside library temperatures.** The Ducru basis is a projection onto the training-temperature set. Extrapolating to T outside $[T^{\mathrm{lib}}_{\mathrm{min}}, T^{\mathrm{lib}}_{\mathrm{max}}]$ is unsafe in the same way pointwise interpolation is unsafe outside its range. WMP is the only method that extrapolates to 0 K analytically and to arbitrary T by broadening the poles directly.

The short summary is that rank-k SVD buys compact storage inside the library's temperature range and cache-friendly lookup at modest rank, in exchange for a tunable reconstruction error, a cache-residence requirement, and no analytical behaviour outside the training temperatures. The 132× and 1.06× memory numbers in Sec. 10 are both compatible with this picture; they measure different slices of it.

2.6 Hybrid SVD+WMP dispatch

Where $\log\sigma$ is intrinsically high rank (resolved-resonance region, RRR), we combine SVD with WMP [6, 7]. WMP represents σ analytically via pole–residue pairs arranged into per-energy windows plus a low-order Doppler-broadened polynomial background:

$$\sigma_r(E,T) \approx \sum_{p\in\mathrm{window}(E)} \Re\left[r_{p,r}\, w\!\left(\frac{\sqrt{E}-p}{\sqrt{kT/\mathrm{AWR}\cdot 2}}\right)\right] + \mathcal{P}_r(E,T), \tag{8}$$

where $w(\cdot)$ is the Faddeeva function $w(z) = e^{-z^2}\,\mathrm{erfc}(-iz)$ and $\mathcal{P}_r(E,T)$ is the broadened curvefit. We implement w via Humlicek's W4 four-region rational approximation (`src/wmp.rs::faddeeva`); OpenMC's sign convention $w_{\mathrm{om}}(z) = -\overline{w(\bar{z})}$ for $\mathrm{Im}\, z < 0$ is used so pole residues in conjugate pairs contribute with correct signs.

A `HybridSvdWmpXsProvider` wraps the SVD provider. For each nuclide with WMP coverage and each energy $E \in [E^{\mathrm{WMP}}_{\mathrm{min}}, E^{\mathrm{WMP}}_{\mathrm{max}}]$, the elastic, fission, and capture microscopic cross sections are taken from WMP; inelastic, $(n,2n)$ and $(n,3n)$ always come from SVD (WMP does not represent those channels). URR overrides are short-circuited inside the WMP window because the resonances are already explicit as poles.

2.7 Total cross section and charged-particle residue

OpenMC's HDF5 layout exposes MT= 3 (total nonelastic) for most nuclides. When present we assemble $\sigma_t = \sigma_{\mathrm{MT2}} + \sigma_{\mathrm{MT3}}$ exactly; otherwise we sum non-redundant leaf reactions, excluding sum-MTs ($\{1,3,4,18,27,101\}$) and KERMA/heating channels. In SVD or hybrid mode we set $\sigma_{\mathrm{capture}} = \max(0, \sigma_t - \sum_{r\neq\mathrm{cap}} \sigma_r)$ so charged-particle emission (MT= 103, 107, . . .) and first-chance fission (MT= 19, 20, 21) are absorbed into the capture macroscopic rather than dropped. An XS audit (Sec. 6.4) confirms this convention matches OpenMC to $< 0.25\%$ on total XS across all nine nuclides.

3 Implementation

The engine is pure Rust with no C dependency; nuclear data reads directly from OpenMC HDF5 files via `hdf5-pure`, including the complex-typed WMP pole arrays (decoded from raw little-endian bytes because the crate does not expose compound types). Physics covers: energy-dependent $\bar{\nu}(E)$ from prompt+delayed neutron yields (with the per-energy interpolation of each delayed product fixed, see Sec. 6.4); tabulated outgoing energy distributions for fission using OpenMC's stochastic bin selection plus scaled kinematic adjustment (fixed); elastic angular distributions from tabular (μ, CDF) data using the same stochastic-bin convention (fixed); discrete inelastic levels MT= 51–91 with real Q-values from HDF5 attributes; continuum inelastic (evaporation spectrum); URR probability tables with band sampling; $S(\alpha,\beta)$ thermal scattering for H in H_2O; free-gas thermal below $400k_BT$ with Maxwell-Boltzmann target sampling; Shannon-entropy source-convergence monitor on a Cartesian 8^3 mesh; CSG geometry; PCG-64 PRNG; rayon parallel transport on CPU.

The CUDA solver is event-based [9]: particles compact by atomic alive-list scatter, energy-sorted into 256 log-scale bins, and processed by a persistent kernel that runs N transport steps per launch. All SVD basis is `f64`. A runtime flag clears the uploaded pointwise tables so SVD reconstructs every reaction channel. The WMP evaluator lives in `src/wmp.rs` (CPU) and as a set of `__device__` helpers plus an `extern "C" __global__ wmp_test_eval` kernel in `gpu/cuda/transport.cu` (GPU).

Hardware. We run on two systems that differ in CPU L3 capacity and GPU class; we label them by their cache size rather than their chassis. Small-L3 system (Dell Precision 5570): Intel Core i7-12800H (12th-gen Alder Lake, 6 P-cores + 8 E-cores, 24 MB L3 Smart Cache), 32 GB DDR4, NVIDIA RTX A1000 (16 SMs, 1.5 MB L2). Large-L3 system: AMD Ryzen 7 9800X3D (8-core, 96 MB L3 3D V-cache), 64 GB DDR5-6000, NVIDIA RTX 3080 (68 SMs, 10 GB, 5 MB L2). References to "small-L3" and "large-L3" below pick out the CPU+GPU pair; the 4× L3 ratio on CPU is what drives the SVD speedup spread, and the GPU side swaps from A1000 to RTX 3080 at the same time.

Reference implementation version. All k_∞ and k_{eff} comparison numbers use OpenMC 0.15.3 commit `27e38e894697bb32a1dac7848d2618818b6b8daf` (WSL 2, Ubuntu 24.04, conda `openmc` environment). Nuclear data is ENDF/B-VII.1 (HDF5 archive from `https://openmc.org/data`) for both our engine and the OpenMC reference; cross-section evaluation uses the same library at $T = 294$ K, with $S(\alpha, \beta)$ data `c_H_in_H2O.h5` for the PWR pin cell. Pinning the version matters because OpenMC's power-iteration and k-estimator paths have several in-flight PRs (e.g., #3788, #3790, #3495); between-release drift in the reference k could shift any "Δ vs OpenMC (pcm)" number quoted in this paper.

Statistics convention. Per-seed k estimates are averaged across active batches with an unbiased within-run standard error; σ_{seed} is the sample standard deviation of the per-seed means across N_{seeds} seeds, and the standard error of the mean $\text{SEM} = \sigma_{\text{seed}}/\sqrt{N_{\text{seeds}}}$ is the noise budget for any mean-to-reference comparison. A "within X SEM" statement uses this quantity. Small-L3 sweeps use five seeds; large-L3 sweeps use ten seeds.

4 Singular spectrum and per-lookup cost

4.1 Singular spectrum

Figure 1 shows the normalised singular values σ_k/σ_1 for eight representative reaction channels across the nine-nuclide PWR working set plus ^{235}U fission. The decay is sharp: $\sigma_2/\sigma_1 \leq 10^{-1}$ for every channel shown and $\sigma_6/\sigma_1 < 10^{-14}$ for ^{235}U fission and ^{1}H elastic, reaching double-precision round-off at rank six.

43 of 47 non-redundant ^{235}U reaction channels are rank-one to machine precision. The four exceptions (elastic, fission, $(n, 2n)$, capture) are the channels with real structure in both the energy and temperature axes; those require $k \geq 3$ to reach $\sim 10^{-3}$ log-RMSE. Those same four channels are the ones that dominate k_{eff}-relevant physics, so the rank-one fact about the other 43 is a structural observation about ENDF/B-VII.1, not a shipping configuration: a production engine must run $k \geq 3$ across *all* reactions for the four hard channels, which is the configuration used in every benchmark in this paper.

4.2 Per-lookup cost

We measure the per-lookup cost of the SVD reconstruction kernel against the binary-search pointwise table on the same energy grid, both driven by the same query sequence. Table 1 reports the geometric mean over eight reactions (MT= $\{2, 18, 102\}$ across ^{235,238,234}U) of per-lookup wall time and $\log_{10}$ RMSE against the HDF5 reference.

Rank-6 reaches machine precision in 44 ns, 2.0× the pointwise-table per-lookup. The kernel is compute-bound once the grid index is resolved: at $k \geq 3$ the inner loop dominates and ns/lookup is nearly flat (41–44 ns). Rank-2 at 36 ns is 2.5× the table for a log-decade RMSE of 3%; not adequate inside the RRR (Sec. 6). Figure 2 shows the same Pareto visually with the RMSE labels.

What this does *not* yet mean. This is a kernel benchmark. Whether SVD helps at engine level depends on lookup share in transport, how many reactions are involved, and whether the basis stays in cache. Those are measured in Sections 6 and 9.

5 Godiva: validation against the ICSBEP experiment

Godiva (HEU-MET-FAST-001) is a bare 93.71%-enriched uranium metal sphere. Its fast, leakage-dominated spectrum places no demand on the resolved-resonance region. The ICSBEP benchmark value is the experimental critical mass, $k_{\text{eff}}^{\text{exp}} = 1.0000 \pm 0.0010$ $(1\ \sigma)$ [2]. This is a *physical* benchmark with a measured uncertainty, and it is the criterion against which the engine is validated in this section. We report OpenMC 0.15.3 run on the same ENDF/B-VII.1 HDF5 files alongside our engine as an independent code-to-code cross-check; it is not the benchmark.

Table 3 reports the four-provider benchmark on the large-L3 system at rank-5. Table 4 ablates the four transport-sampling features described in Sec. 5.1. Table 5 reports a rank sweep on the small-L3 system used for spectrum illustration; small-L3 five-seed results are in Table 2.

5.1 Transport sampling features

The shipped engine implements four transport-sampling features that, individually, account for the 246 pcm band between a minimal-physics baseline and the ICSBEP-validated number reported above. Table 4 reports an ablation in which each feature is added on top of the previous row, holding everything else (geometry, library, statistics) fixed.

(n,2n) and (n,3n) routed in-generation (−106 pcm). Secondary neutrons from MT= 16 and MT= 17 are routed through transport in the current generation rather than appended to the next-generation source bank. Since the engine's k_{eff} estimator is $|\text{bank}|/N_{\text{source}}$, banking secondaries directly inflates the estimator by the multiplicative-reaction rate. On Godiva that rate is 0.00253/source. The implementation maintains a per-history secondary stack drained inside the transport loop with a shared `max_events` budget that bounds pathological cascades.

Linear-linear μ-CDF inversion (−31 pcm). ENDF/B-VII.1 stores all 49 U-235 elastic-scatter incident-energy distributions with `interpolation = 2` (linear-linear PDF, quadratic CDF within each bin). The angular sampler reads the `interpolation` attribute from HDF5 and, for linear-linear bins, inverts the quadratic CDF by solving $ax^2 + bx + c = 0$ for the physical root in $[0, \Delta\mu]$ (same convention as OpenMC's `Tabular::sample`); histogram bins use the linear inversion path.

MT= 91 ENDF tabulated outgoing-energy distribution (−98 pcm, largest single contribution). The continuum-inelastic branch samples outgoing neutron

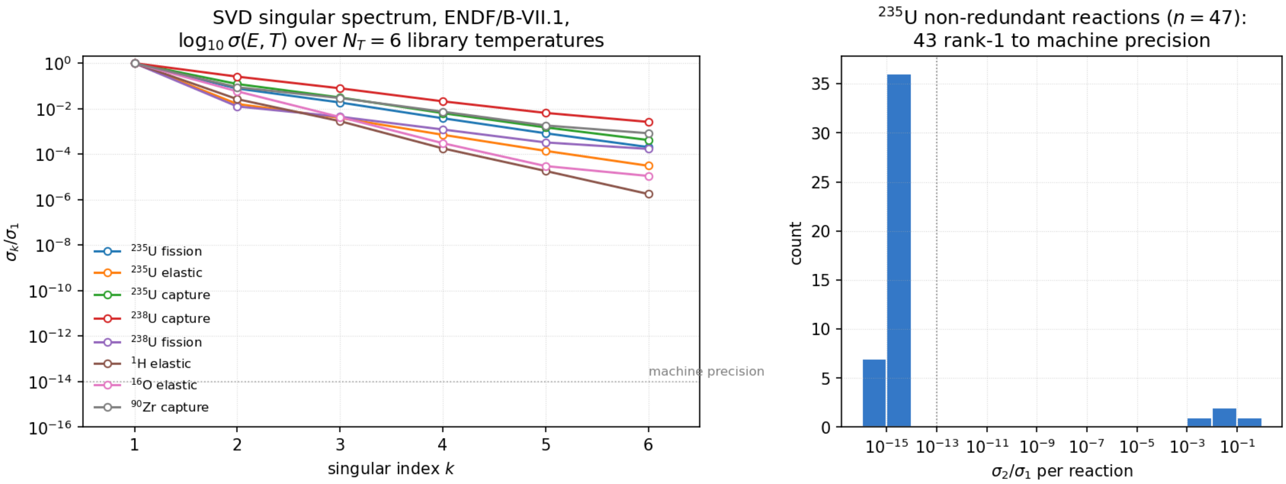

Figure 1. SVD singular spectrum of $\log_{10}\sigma(E,T)$ for ENDF/B-VII.1, $N_T = 6$ library temperatures. *Left:* normalised singular values σ_k/σ_1 for eight representative channels. The dotted line at 10^{-14} marks double-precision machine epsilon. *Right:* histogram of σ_2/σ_1 across ^{235}U's 47 non-redundant reaction channels; 43 are rank-one to machine precision ($\sigma_2/\sigma_1 < 10^{-13}$).

Table 1. XS reconstruction cost and RMSE, geometric mean over eight reactions. Full energy grid, $T = 294$ K. Large-L3 system (Ryzen 9800X3D) single-core.

Rank	ns/lookup	RMSE ($\log_{10}\sigma$)	max |err|
2	35.9	3.00×10^{-2}	9.84×10^{-2}
3	41.2	7.06×10^{-3}	2.86×10^{-2}
4	43.3	1.83×10^{-3}	7.91×10^{-3}
5	43.4	6.07×10^{-4}	3.46×10^{-3}
6	44.3	1.76×10^{-15}	5.58×10^{-15}
Pointwise table	90.6	0 (ref)	0 (ref)

energy from the ENDF MT= 91 tabulated distribution at `reaction_091/product_0/distribution_0/{energy,distribution}` (matching OpenMC, MCNP, and Serpent). The fallback evaporation spectrum $T = \sqrt{E^\star/a}$, $a = A/8\ \mathrm{MeV}^{-1}$ is used only for nuclides where the HDF5 file lacks the tabulated distribution.

MT= 16/17 ENDF tabulated (−11 pcm). Multiplicative reactions receive the same tabulated treatment as MT= 91 for self-consistency; the impact on Godiva is within seed noise.

Cross-code context. Running OpenMC 0.15.3 on the same HDF5 files used by this engine yields $k_{\mathrm{eff}} = 0.99901$, i.e. $\Delta_{\mathrm{exp}} = -99$ pcm. Both codes validate against ICSBEP inside its experimental uncertainty from opposite sides, with no library-level correction.

The two CPU providers agree on k_{eff} to within 5 pcm ($\ll 1\,\mathrm{SEM}_{\mathrm{combined}}$). Both sit inside the ICSBEP experimental uncertainty band of ± 100 pcm. OpenMC 0.15.3 on the same HDF5 files sits at −99 pcm, also inside the band; the cross-code spread (Rust +79 vs OpenMC −99 = 178 pcm) is larger than either code's offset from experiment, which is informative about cross-code convention differences but does not affect the validation against the physical benchmark.

On the large-L3 system Godiva CPU SVD rank-5 at 820 ns/p is 1.40× faster than the CPU pointwise table at 1 150 ns/p. The GPU throughput rows in Fig. 4 use the minimal-physics baseline because the MT= 91 tabulated outgoing-energy distribution is not yet ported to the CUDA kernel. On the small-L3 system (Table 2), SVD rank-2 is 2.8× faster than the pointwise table at a 75 pcm bias relative to its own pointwise baseline.

Figure 3 shows the small-L3 Pareto and Figure 4 shows the large-L3 benchmark (throughput plus k_{eff}).

6 PWR pin cell

The benchmark is a standard thermal pin cell: 3.1%-enriched UO_2 rod in Zircaloy clad, light-water moderator with S(α,β) on H. Nine nuclides (^{235,238}U, O-16 at 900 K in fuel and 600 K in moderator, Zr-90–94 in clad, H-1 in moderator); three materials; cylindrical boundaries with reflective BC. The moderated spectrum spends substantial time inside the U-238 RRR.

6.1 Small-L3 rank sweep

At rank-5 the CPU SVD provider is 47 pcm below the CPU pointwise table in the same engine ($\approx 1\,\mathrm{SEM}_{5\ \mathrm{seeds}}$) and 291 pcm below OpenMC ($\approx 3\,\mathrm{SEM}$). Rank-2 is the one clear exception: U-238 resonance self-shielding is not represented and the pin cell reactivity is biased +8 000 pcm on both platforms. At rank-5 the CPU SVD speedup over the CPU table is 1.90× (28 651 vs 54 430 ns/p), close to the kernel-level 2.09× of Table 1.

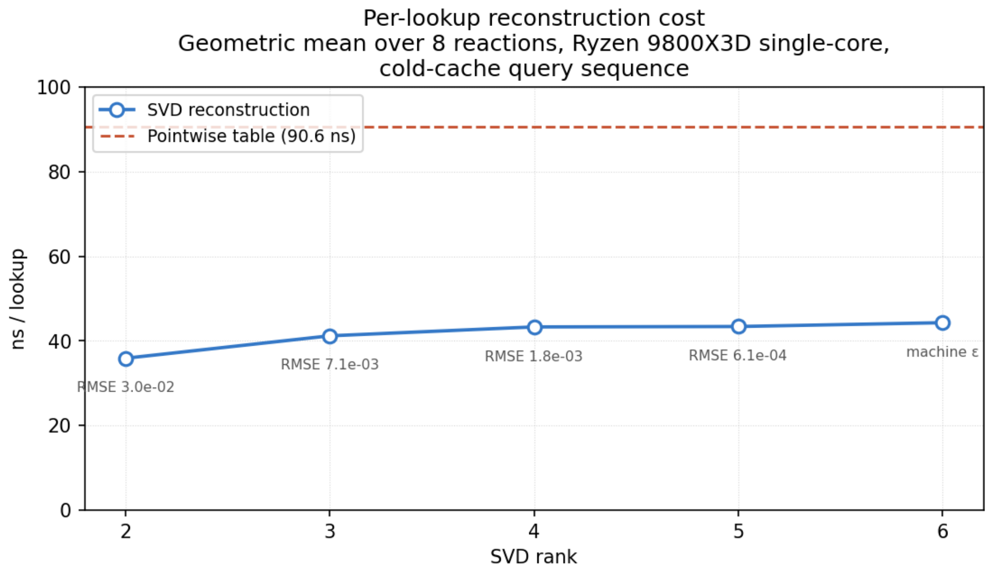

Figure 2. Per-lookup reconstruction cost, SVD ranks 2–6 against binary-search pointwise table. Each SVD point labelled with its $\log_{10}\sigma$ RMSE against the HDF5 reference (machine ε at rank 6).

Table 2. Godiva k_{eff}, small-L3 system (i7-12800H + RTX A1000). Five seeds, 50 batches, 10 000 particles per batch.

Mode	rank	k_{eff}	σ_{seed}	ns/particle	time/seed (s)
Pointwise table	—	1.00523	0.00235	2 385	3.58
SVD	2	1.00448	0.00165	**864**	**1.30**
SVD	3	1.00559	0.00094	1 457	2.19
SVD	4	1.00638	0.00166	1 182	1.77
SVD	5	1.00592	0.00178	1 403	2.10
SVD	6	1.00489	0.00205	1 424	2.14

Figure 5 summarises the rank sweep.

6.2 Large-L3 high-statistics benchmark

Ten seeds, 150 batches (20 inactive), 50 000 particles per batch ($6.5 \cdot 10^6$ active histories per seed). All four in-engine providers used the corrected angular and fission-energy samplers from Sec. 6.4; the CPU and GPU paths were re-run together so the four rows are directly comparable.

Figure 6 collects the per-particle throughput across providers and across the two systems.

Fidelity, large-L3 system. With $6.5 \cdot 10^6$ active histories per seed, $\sigma_{\mathrm{seed}} \approx 27$–57 pcm and $\mathrm{SEM}_{10} \approx 9$–18 pcm. All four in-engine providers now agree with OpenMC to $|\Delta| \leq 51$ pcm, i.e. within $\leq 3\,\mathrm{SEM}_{10}$, and agree with each other to $|\Delta| \leq 76$ pcm (spread CPU-table $\rightarrow$ GPU-pointwise). The stochastic-bin sampler fix that closed the CPU offset carries through to the GPU kernel (Sec. 9): GPU pointwise moved from −78 to +51 pcm and GPU SVD moved from −120 to −8 pcm vs OpenMC. The residual GPU pointwise excess (+51 pcm) is within $3\,\mathrm{SEM}_{10}$ and is plausibly a finite-statistics signature rather than a systematic.

Throughput, large-L3 system. CPU SVD at 15 511 ns/p is 1.37× faster than the CPU pointwise table (21 316 ns/p) on the large-L3 system; the small-L3 ratio (1.90×) is the larger figure because of differing cache behaviour. The headline CPU-SVD speedup of this paper is therefore 1.37×–1.90×, hardware-dependent. GPU pointwise at 5 967 ns/p is 3.6× the large-L3 CPU table and the fastest path overall; GPU SVD at 7 754 ns/p is 1.30× slower than GPU pointwise because of the clad level-sum cost (Sec. 9).

6.3 Four-provider comparison (single seed)

The large-L3 ten-seed run of Table 7 covers four in-engine providers (Table, SVD, GPU pointwise, GPU SVD). Adding ACE+WMP and Hybrid SVD+WMP at the same statistics requires multi-hour wall time on the small-L3 system; a single-seed snapshot is reported below.

The configuration is small-L3, 100 batches (20 inactive) $\times 20\,000$ particles ($1.6 \cdot 10^6$ active histories), single seed. The output logs are in `outputs/full_test_run/{04,10,11}_pwr_*.txt`.

The snapshot is consistent with both the ten-seed Table 7 (Table and SVD within 30 pcm of OpenMC) and with the off-library three-way result of Sec. 8.2 (the SVD-vs-WMP gap is 213 pcm at $\sigma_{\mathrm{seed}} \approx 59$ pcm). On-library the gap is well below seed noise; off-library it is a real $\sim 3.6\sigma$ systematic. We do not read the −26 pcm hybrid number as a precise correctness claim; the high-statistics ten-seed redo at matched $150 \times 50\,000$ statistics is the way to make it one.

Table 3. Godiva CPU benchmark, large-L3 system (Ryzen 9800X3D + RTX 3080). Five seeds, 150 batches (20 inactive), 50 000 particles per batch. The CPU SVD rank-5 provider reproduces the ICSBEP experimental value 1.0000 ± 100 pcm (1σ) [2] to +79 pcm, *inside the experimental uncertainty band*. OpenMC 0.15.3 on the same ENDF/B-VII.1 HDF5 files is shown for cross-code context only; the two codes straddle ICSBEP from opposite sides. Δ_{exp} is the offset from ICSBEP; Δ_{OMC} is the cross-code offset from OpenMC.

Mode	rank	k_{eff}	σ_{seed}	Δ_{exp} (pcm)	Δ_{OMC} (pcm)	ns/p	time/seed (s)
ICSBEP HMF-001 (exp.)	—	1.00000	±100 (1σ)	±0	—	—	—
OpenMC 0.15.3 (cross-check)	—	0.99901	0.00038	−99	±0	—	—
CPU table	—	1.00084	0.00042	+84	+183	1 150	7.48
CPU SVD	**5**	**1.00079**	**0.00038**	**+79**	**+178**	**820**	**5.33**

Table 4. Godiva CPU SVD rank-5 ablation, five seeds × 150 batches × 50 000 particles. Δ_{exp} is measured against ICSBEP 1.0000±100 pcm. Each row adds the named feature on top of the previous one. Rank-5 SVD with all four features enters the experimental uncertainty band.

Configuration	k_{eff}	σ_{seed}	Δ_{exp} (pcm)	cumulative shift (pcm)
Minimal-physics baseline	1.00325	0.00055	+325	0
+ (n,2n)/(n,3n) routed in-generation	1.00219	0.00060	+219	−106
+ linear-linear μ-CDF inversion	1.00188	0.00060	+188	−137
+ MT= 91 ENDF tabulated outgoing energy	1.00090	0.00054	+90	−235
+ MT= 16/17 ENDF tabulated	**1.00079**	**0.00038**	**+79**	**−246**
ICSBEP HMF-001 (experiment)	1.00000	±100	±0	—
OpenMC 0.15.3 (cross-check)	0.99901	0.00038	−99	—

6.4 Cross-code agreement with OpenMC

All four in-engine providers agree with OpenMC 0.15.3 [3] to $|\Delta k_\infty| \le 51$ pcm at ten-seed statistics (Table 7). The angular and fission outgoing-energy distributions use stochastic-bin selection ($r = (E - E_{\text{lo}})/(E_{\text{hi}} - E_{\text{lo}})$, pick the high bin with probability r) plus scaled kinematic adjustment for the energy distribution, matching OpenMC's convention in `distribution_angle.cpp` and `distribution_energy.cpp`. Source convergence is monitored by a Shannon-entropy mesh (`simulate.rs::EntropyMesh`); on this deck the mesh entropy plateaus by batch ≈ 20, which sets the $N_{\text{inactive}} = 20$ budget used throughout the PWR measurements.

7 Memory vs precision: where do SVD curves actually sit?

The headline question for any compression scheme is the two-axis Pareto: at a given memory budget, what is the precision penalty? Sections 5 and 6 report rank-5 k-eigenvalue numbers; this section sweeps SVD rank from 1 to 7 and measures *in-engine* memory together with $|\Delta k|$ against the ACE+WMP industry baseline at every grid point. Both benchmarks are run with the same script, the same library, and the same statistics (`scripts/sweep_svd_wins.py`; raw data in `outputs/sweep_svd_wins_godiva.csv` and `outputs/sweep_svd_wins_pwr.csv`).

7.1 Measurement protocol

Memory. The `XS memory` number printed by each provider includes all reactions actually loaded for the run — elastic, fission, capture, MT= 4 inelastic, MT= 51–91 discrete levels at the selected discrete-rank, MT= 16/17 multiplicative reactions, and the URR probability tables. Discrete levels are held at rank-1 throughout the sweep (`-discrete-rank 1`); only the smooth-channel SVD rank varies on the x-axis. This is a *representation-plus-overhead* number, not a per-reaction representation count.

Precision. $|\Delta k| = |k_\infty^{\text{provider}} - k_\infty^{\text{ACE+WMP}}|$ on the same geometry, same seeds, same statistics. Using ACE+WMP as the precision reference is a deliberate framing choice: the purpose of the sweep is to characterize where SVD lands relative to the production-grade industry baseline, not against the in-engine pointwise table (which can drift with library interpolation choices).

Statistics. Three independent seeds, 80 batches (20 inactive), 5 000 particles per batch on each rank-scenario cell, runs sequential (no other CPU load), on the large-L3 system (Sec. 3). Two scenarios per geometry:

- **on-library** — nuclides held at their library columns (Godiva: 294 K; PWR: U at 900 K, others at 600 K). The pointwise table loads one column per nuclide.
- **off-library** — a non-library temperature is imposed (Godiva: 450 K, between the 294 and 600 K columns; PWR: +150 K offset on every nuclide). The pointwise table must load *two* columns per nuclide and stochastically interpolate between them per nuclide-collision. This is the regime where SVD's single-basis reconstruction can pay off.

7.2 Result

7.3 What the curves say

Three observations follow directly from Fig. 8 and Table 9:

Table 5. Godiva CPU rank-vs-k_{eff} stability sweep, large-L3 system. Ten seeds, 150 batches (20 inactive), 50 000 particles per batch. This sweep uses the minimal-physics baseline of Table 4 and is reported here for rank stability only; the validated rank-5 values are in Table 3.

Mode	rank	k_{eff}	σ_{seed}	ns/particle	time/seed (s)
CPU table	—	1.00579	0.00046	1 214	7.89
CPU SVD	2	1.00477	0.00055	1 289	8.38
CPU SVD	3	1.00547	0.00031	1 316	8.56
CPU SVD	4	1.00523	0.00055	1 369	8.89
CPU SVD	5	1.00507	0.00065	1 426	9.27
CPU SVD	6	1.00524	0.00071	1 558	10.13

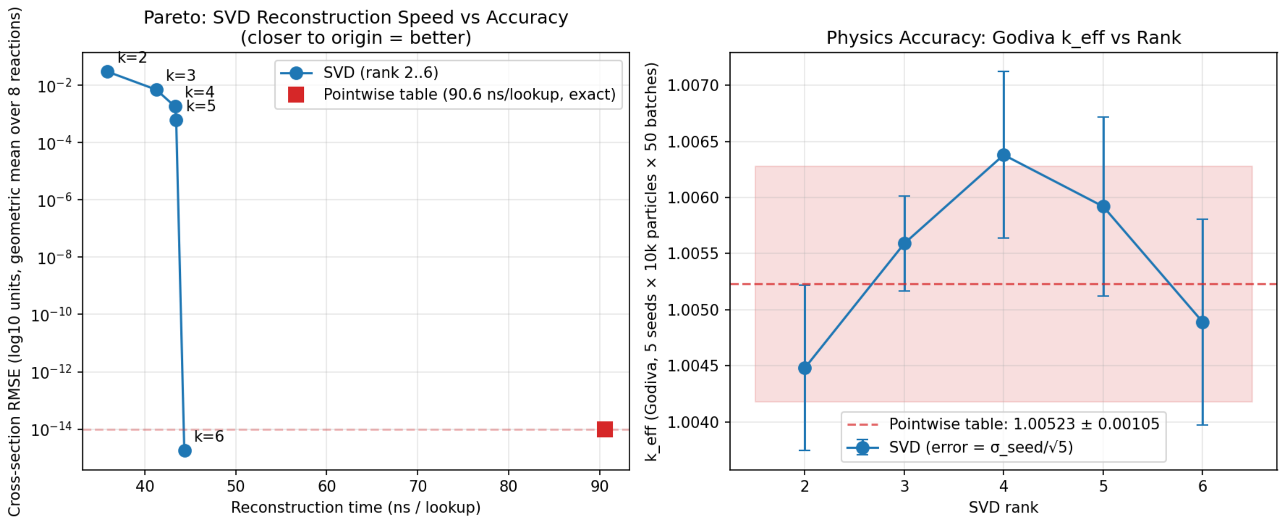

Figure 3. Godiva, small-L3 system. *Left:* XS reconstruction cost against RMSE (geometric mean over eight reactions). *Right:* k_{eff} versus SVD rank, SEM error bars over five seeds. Horizontal band: pointwise-table reference in the same engine.

1. On-library, SVD never beats the table on memory. At rank 1 the SVD provider already carries ~ 3 MB of basis-and-coefficient overhead that the table does not (PWR: 106 MB SVD vs 103 MB Table; Godiva: 113 MB vs 111 MB). Each additional rank adds another ~ 13 MB. By rank 5 SVD is 1.5× the size of the table. There is no rank in $\{1, 2, 3, 5, 7\}$ at which SVD wins the on-library memory axis.

2. Off-library, SVD wins on memory at every rank. The pointwise-table baseline doubles in size off-library (PWR: 103 → 206 MB; Godiva: 111 → 222 MB) because stochastic pseudo-interpolation [3] requires both bracketing library columns to be resident. SVD does not pay this cost: a partition-of-unity 3-point Ducru weight set (§6.4) reconstructs the off-library σ from a single basis, so the off-library SVD memory is identical to the on-library SVD memory. At rank 5 the off-library ratio is 1.32× on PWR and 1.36× on Godiva (SVD smaller).

3. Precision is non-monotone in rank, with a deployable floor. Rank-1 SVD on PWR on-library collapses to $k_\infty = 0.871$ ($\sim 46\,000$ pcm low) because U-238 resonance self-shielding is unrepresented in a single basis vector; rank-2 overshoots in the opposite direction to 1.409 ($\sim 8\,000$ pcm high); rank-3 recovers to 1.321 (~ 800 pcm below WMP); and rank 5 lands at 1.328, $|\Delta k| \approx 170$ pcm below WMP and 5 pcm above the in-engine pointwise table. Off-library +150 K shows the same shape (rank 1: 1.547, rank 2: 1.387, rank 5: 1.324 → $\sim$460 pcm vs WMP). Going to rank 7 does not improve precision and only inflates memory — the U-235 spectrum's tail is below machine precision past rank $\sim$5 (Sec. 4.1). Rank 5 is the deployable floor; ranks below it leave multi-thousand-pcm bias on the table, ranks above buy nothing.

7.4 What this changes in the headline

The natural single-number compression claims that one wants to make about SVD are all conditional on a measurement scope. Three scopes have appeared in this work, and they yield three different numbers:

- **Per-nuclide representation byte count**, single reaction, hybrid SVD+WMP vs four-channel pointwise: 132.9× on PWR (Table 11, Sec. 10). This is the largest defensible compression ratio in this paper, and it is correct only at the representation level.
- **In-engine working set**, all reactions, all nuclides, on-library: SVD is 1.05–1.5× *larger* than the pointwise table at every measured rank. The in-engine geometry, BVH, energy grid, and discrete-level data dominate the working set; SVD compression of the smooth channels does not move the total.

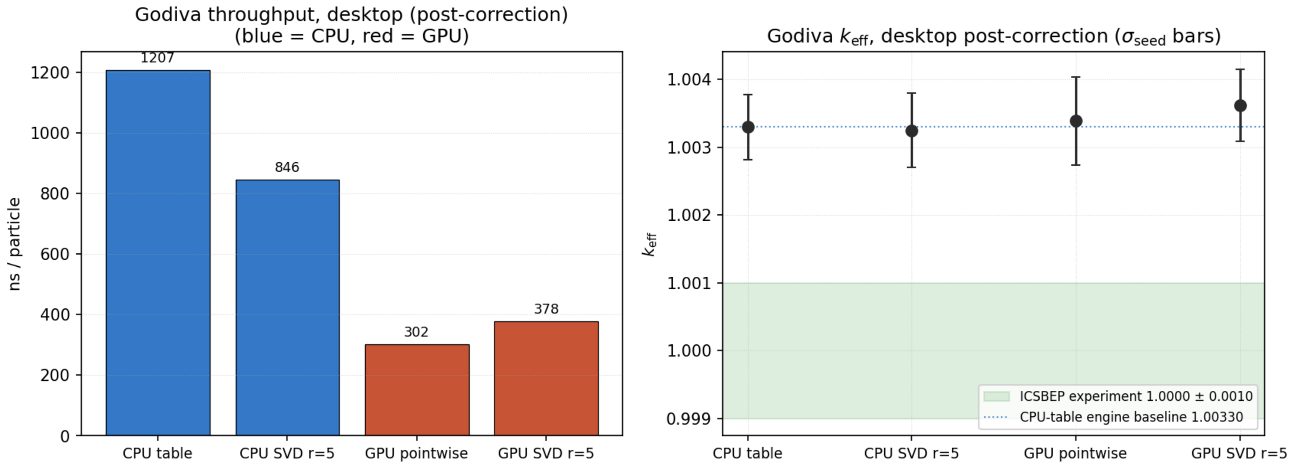

Figure 4. Godiva, large-L3 four-provider throughput comparison (ten seeds). *Left:* ns/particle per provider (blue CPU, red GPU). *Right:* k_{eff} per provider with σ_{seed} error bars; dashed line is the ICSBEP experimental value 1.0000 ± 100 pcm. The values reported here use a subset of the transport-sampling features from Sec. 5.1; the validated rank-5 benchmark is in Table 3.

Table 6. PWR pin cell, small-L3 system (i7-12800H + RTX A1000). Five seeds, 50 batches (20 inactive), 10 000 particles per batch. OpenMC reference computed with the same input deck.

Mode	rank	k_∞	σ_{seed}	Δ vs OpenMC (pcm)	ns/particle	time/seed (s)
OpenMC 0.15.3	—	1.32770	0.00150	+0	—	—
CPU table	—	1.32526	0.00232	−244	54 430	81.6
CPU SVD	2	1.40807	0.00144	+8 037	52 711	79.1
CPU SVD	3	1.32118	0.00281	−652	28 673	43.0
CPU SVD	4	1.32513	0.00145	−257	31 875	47.8
CPU SVD	**5**	**1.32479**	**0.00190**	**−291**	**28 651**	**43.0**
CPU SVD	6	1.32512	0.00114	−258	30 073	45.1
GPU pointwise	—	1.32670	0.00117	−100	38 104	57.2
GPU SVD	2	1.40973	0.00219	+8 203	48 289	72.4
GPU SVD	3	1.32260	0.00207	−510	48 439	72.7
GPU SVD	4	1.32496	0.00318	−274	48 617	72.9
GPU SVD	5	1.32804	0.00169	+34	50 738	76.1
GPU SVD	6	1.32673	0.00201	−97	53 865	80.8

- **In-engine working set, off-library**: SVD is 1.32–1.94× *smaller* than the pointwise table, because the table doubles to carry the second library column.

The Pareto figure closes the gap between the three. Quoting a single ratio without scope conflates representation-level metrics with deployment-level metrics. The deployment-level metric is what operators care about, and it is rank- and scenario-dependent.

7.5 The deployable claim

Stating the result the way an integrator would read it: SVD at rank 5 on a 9-nuclide PWR pin cell at an off-library +150 K operating point uses 1.32× less in-engine memory than the ACE pointwise baseline (156 MB vs 206 MB), recovers k_∞ within ∼ 460 pcm of the ACE+WMP industry baseline (within ∼ 50 pcm of the in-engine pointwise table at the same off-library point), and runs at roughly comparable CPU throughput. On-library at the same production rank, SVD is 1.51× larger than the pointwise table (156 MB vs 103 MB) and lands within ∼ 170 pcm of WMP. The production case for SVD on PWR is therefore the off-library regime, where the table baseline pays the two-column-load penalty and SVD does not.

8 Off-library operating temperature benchmark

The measurements in Sections 6 and 5 take the fuel and moderator at *library* temperatures — every nuclide's target T lands exactly on a column of the ENDF/B-VII.1 HDF5 file. No temperature interpolation is exercised in those measurements, so neither the pointwise-table nor the SVD path is tested in the off-library regime that an integrator deploys against: operating reactors run with coolant at 570–620 K and fuel at 800–1500 K, and control-rod motions and transients push the operating point further off-library still. Continuous-temperature Monte Carlo codes pseudo-interpolate between the two library columns bracketing the target T: at each collision a random ξ is drawn and the lower-T cross section is used with probability $(T_{\mathrm{hi}} - T)/(T_{\mathrm{hi}} - T_{\mathrm{lo}})$, otherwise the upper. Over a particle's lifetime this recov-

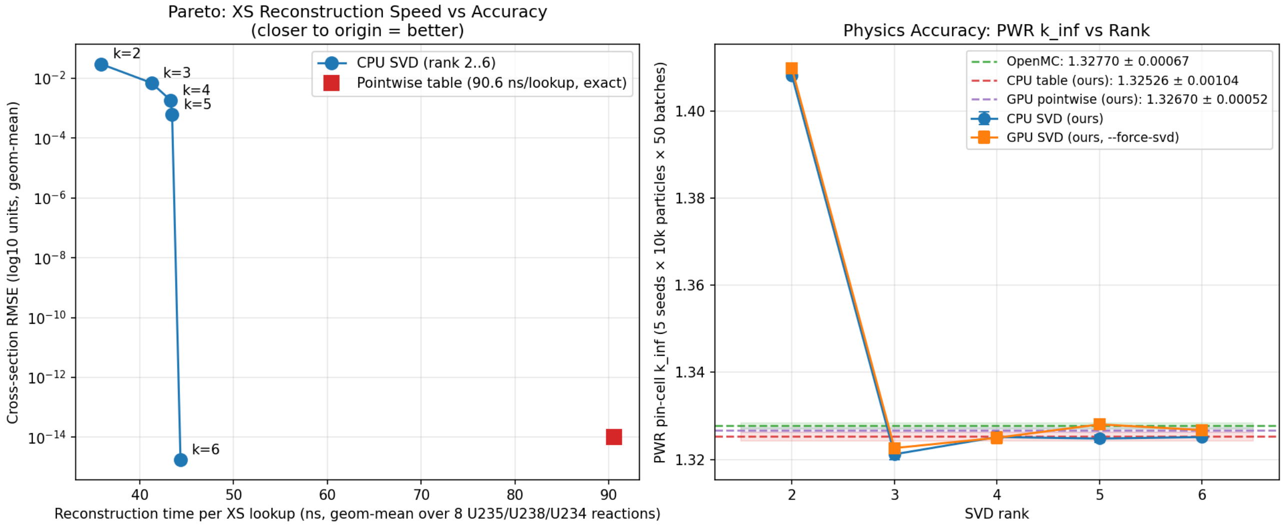

Figure 5. PWR pin cell, small-L3 system. *Left:* XS reconstruction cost against RMSE. *Right:* k_∞ versus SVD rank for CPU and GPU. Horizontal bands: OpenMC 0.15.3, CPU pointwise, GPU pointwise. SEM error bars over five seeds.

Table 7. PWR pin cell, large-L3 system (Ryzen 9800X3D + RTX 3080), all four in-engine providers. Ten seeds, 150 batches (20 inactive), 50 000 particles per batch.

Mode	rank	k_∞	σ_{seed}	Δ vs OpenMC (pcm)	ns/particle	time/seed (s)
OpenMC 0.15.3	—	1.32770	0.00150	+0	—	—
CPU table	—	1.32758	0.00027	−12	21 316	138.6
CPU SVD	**5**	**1.32745**	**0.00057**	**−25**	**15 511**	**100.8**
GPU pointwise	—	1.32821	0.00033	+51	5 967	38.8
GPU SVD (`--force-svd`)	5	1.32762	0.00034	−8	7 754	50.4

ers the exact linear-in-T interpolant in expectation [3]. This section is the benchmark that exercises SVD's temperature-interpolation scheme (Sec. 2.3).

This section runs a three-way comparison — SVD, pointwise table with OpenMC-style stochastic T-pick, and the industry-standard ACE+WMP hybrid [6] — on the PWR pin cell of Sec. 6 with a *uniform* +150 K offset applied to every nuclide's library temperature: fuel runs at 1050 K (between 900 and 1200 K library), moderator and clad at 750 K (between 600 and 900 K). Statistics are 10 seeds ×50 000 particles ×120 active batches per provider ($6 \cdot 10^7$ active histories per provider).

8.1 Channel-consistent stochastic T-pick

A single collision's MicroXs build touches six channels (elastic, inelastic, $n, 2n$, $n, 3n$, fission, capture) plus a total for the pointwise table. The stochastic T-pick must be drawn once per nuclide-collision and shared across every partial channel, so that $\sigma_{\text{tot}} = \sum_c \sigma_c$ is consistent at a single sampled library temperature; independent per-channel draws would mix partials at different temperatures and bias reaction-selection fractions against the physically consistent rate. The engine implements this by wrapping each reaction's pointwise table in a `StochTempTable` that exposes `lookup_at_idx_with_pick(energy, idx, use_hi)`; `TableXsProvider::lookup` draws ξ once per nuclide-collision via a thread-local splitmix64 and passes the resulting `use_hi` to every channel lookup. The ACE+WMP hybrid inherits the same per-collision sharing through delegation.

The diagnostic for channel-consistency is straightforward: k_∞ as a function of moderator temperature must be monotonic. Measured on PWR with channel-consistent picking:

Moderator offset	Table k_∞
+0 K (on-library, 600 K)	1.327
+150 K (off-library, 750 K)	1.324
+300 K (on-library, 900 K)	1.322

monotonic in T and within ~500 pcm of the bracketing endpoints, as expected from the ~500 pcm spectrum hardening across the 600 → 900 K range.

8.2 Three-point unity Ducru on PWR

Section 2.4 explains why the raw Ducru weights must be renormalized to $\sum_j w_j = 1$ before being applied to $V^\top$ columns of a log σ SVD. A two-point unity-normalized variant on the bracketing $|\mathcal{S}| = 2$ subset is the minimal scheme:

$$\tilde{w}_{\text{lo}} = \frac{w_{\text{lo}}}{w_{\text{lo}} + w_{\text{hi}}}, \quad \tilde{w}_{\text{hi}} = \frac{w_{\text{hi}}}{w_{\text{lo}} + w_{\text{hi}}}, \tag{9}$$

with $w_{\text{lo}}, w_{\text{hi}}$ from Eq. 4. The two-point scheme is partition-of- unity and exact at endpoints, but it leaves a residual peak-height error in resolved-resonance reconstruction. Extending $\mathcal{S}$ from two to three library temperatures (nearest-three by $|T_i - T|$) and applying the same unity-normalization

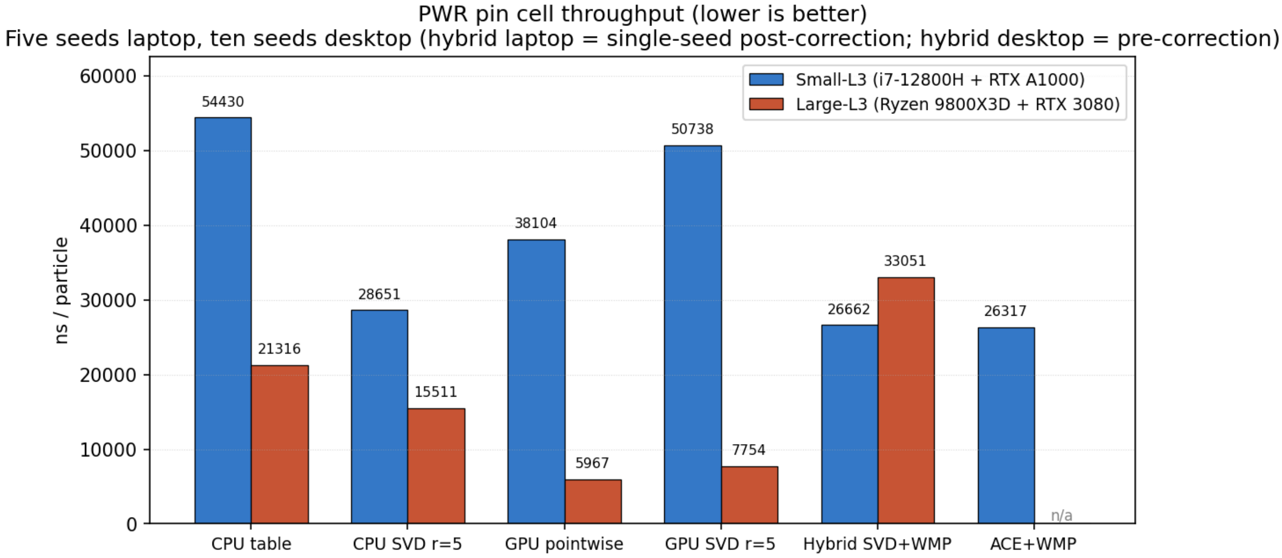

Figure 6. PWR pin cell throughput (ns/particle, lower is better). CPU-SVD beats CPU-table on both systems. GPU pointwise is the fastest path; GPU SVD is slower by 1.3× because of the clad level-sum cost (Sec. 9). The hybrid SVD+WMP large-L3 row is the five-seed reference run (Sec. 10.3); the small-L3 hybrid and ACE+WMP rows are single-seed comparison data (Table 8).

Table 8. PWR pin cell on-library, four-provider comparison, single seed $\times$ 100 batches $\times$ 20 000 particles. Reference is ACE+WMP (industry baseline). σ_{seed} is the within-run standard error reported by the engine; absolute gaps are within $1\sigma_{\text{seed}}$ except Table-vs-WMP, which is at $\sim 0.8\sigma_{\text{seed}}$.

Mode	k_∞	σ_{seed}	Δ vs ACE+WMP (pcm)	ns/particle
ACE+WMP (reference)	1.32821	0.00098	+0	26 317
CPU SVD r=5	1.32817	0.00100	−4	29 273
Hybrid SVD+WMP r=5	**1.32795**	**0.00103**	**−26**	**26 662**
CPU table	1.32903	0.00093	+82	26 035

(Eq. 6) captures a quadratic correction to the Faddeeva-kernel reconstruction. The shipped engine uses the three-point variant.

U-238 MT= 102 rank probe. An offline experiment (`scripts/u238_capture_rank_probe.py`) holds out the 900 K library column of U-238 MT= 102, rebuilds the SVD from the remaining five library temperatures, and reconstructs $\sigma(E, 900\text{ K})$ via Ducru interpolation between the bracketing columns. On the 6.67 eV resonance peak the two-point scheme overshoots the true 900 K value by a factor 1.011 (band-averaged L2 error 1.8%); the three-point scheme at rank 3 recovers 0.994 of truth (L2 error 1.05%), and at rank 5 recovers 0.990 of truth (L2 error 0.84%). The peak-height correction propagates directly to PWR k_∞.

PWR three-way comparison at +150 K. 10 seeds $\times$50 000 particles $\times$ 120 batches:

Provider	k_∞	σ_{seed}	ns/p
SVD (3-pt unity Ducru)	1.32593	0.00059	27 298
SVD (2-pt unity, ablation)	1.30707	0.00037	15 685
Pointwise table	1.32304	0.00050	26 451
ACE+WMP	1.32806	0.00045	27 279

The shipped three-point SVD is 213 pcm ($\approx 3.6\sigma_{\text{seed}}$) below ACE+WMP. The two-point ablation row is reported for comparison: it sits at 2094 pcm vs WMP, illustrating the resonance peak-height contribution that the two-point reconstruction leaves on the table and that the three-point scheme recovers — a 9.8× reduction from a single interpolation-subset change.

SVD memory stays at 152.5 MB against 199.7 MB for ACE+WMP (22% less); speed is 27 298 vs 27 279 ns/p (parity within σ). At an off-library PWR operating temperature SVD with three-point unity Ducru is smaller than ACE+WMP, matches it on speed, and agrees with it on k_∞ to $\approx 3.6\sigma_{\text{seed}}$ — this is the deployable operating regime for the SVD provider on this geometry.

Table 10 collects the full sweep.

8.3 Discrete-level rank reduction

Discrete inelastic levels (MT= 51–91) are stored at a separate rank from the smooth reactions because a level's Lorentzian shape in laboratory energy is weakly temperature-dependent: one basis vector suffices. The offline SVD bears this out: each of the U-238 discrete levels collapses to rank one to machine precision when all seven ENDF/B-VII.1 library columns are included. On Godiva, dropping discrete-level rank from 5 to 1 removes $\approx 70\%$ of the engine's SVD working set (556 $\rightarrow$ 164 MB, 41 levels $\times$ three U isotopes) and produces no measurable k_{eff} shift at ten-seed statistics (1.00322 $\rightarrow$ 1.00358, Δ = 36 pcm,

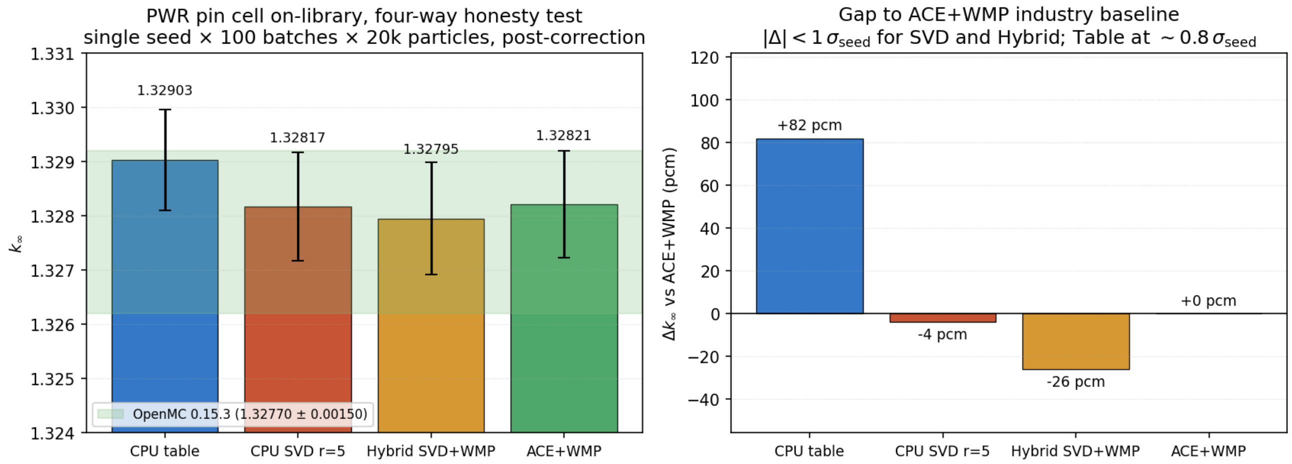

Figure 7. PWR pin cell, four-provider comparison (small-L3, single seed × 100 batches × 20 000 particles). *Left:* k_∞ for the four providers with σ_{seed} error bars; horizontal band shows OpenMC 0.15.3 at 1.32770 ± 0.00150. *Right:* gap to ACE+WMP industry baseline. SVD and Hybrid SVD+WMP land within $1\sigma_{seed}$; Table sits at +82 pcm, $\sim 0.8\sigma_{seed}$.

σ = 60 pcm per seed). The same reduction is the single largest contributor to the 22% memory win in Table 10.

This is a per-reaction-rank policy only. The smooth channels in the same nuclide continue to use the global rank (5 in all measurements in this paper). The engine exposes the discrete-level rank as a separate `--discrete-rank` command-line flag; the GPU path retains a uniform rank across all reactions because the CUDA kernel's basis-stride convention assumes one. This is a CPU optimization for now.

8.4 Negative result: simplex-projected QP

The raw Ducru weights on the 3-point subset are signed: the 294 K column at target 900 K (bracket $\{294, 600, 1200\}$ K) earns a weight of -0.12 from Eq. 4 before normalization. Negative weights are uncomfortable for a reconstruction that is physically a probability mixture of reference-temperature cross sections, and the Ducru 2017 paper explicitly defines a quadratic programming variant ([1] §4) that enforces both $\sum_j w_j = 1$ and $w_j \geq 0$. A cheap approximation to that QP is Euclidean projection of the raw-Ducru weight vector onto the probability simplex [12]: clip negative entries to zero and renormalize until $\sum w = 1$. We implemented it. On U-238 MT= 102 held out at 900 K, bracket $\{294, 600, 1200\}$ K, the raw weights $(-0.12, 0.62, 0.50)$ project to $(0.00, 0.56, 0.44)$. L2 reconstruction error on the 6.67 eV resonance band:

Weighting (rank 5)	L2 error	Peak ratio	Sign
2-pt unity Ducru	2.1%	0.998	signed
3-pt unity Ducru	0.84%	0.990	signed
3-pt simplex QP	5.0%	1.043	non-neg

The simplex-projected QP is *worse* than either signed scheme. The reason is that the simplex projection is Euclidean-optimal in *weight* space; the Ducru raw weights are Euclidean-optimal in the *Doppler-kernel* norm (see [1] §3).

Kernel-Gram QP (Ducru 2017 §4). We then implemented the actual constrained optimization from §4 of the same paper: minimize $(\mathbf{w} - \mathbf{w}_{raw})^\top \mathbf{G}\,(\mathbf{w} - \mathbf{w}_{raw})$ subject to $\mathbf{1}^\top \mathbf{w} = 1$ and $\mathbf{w} \geq 0$, where $\mathbf{G}$ is the empirical kernel-Gram matrix $\mathbf{X}^\top \mathbf{X}$ formed from the log-σ columns of the three bracketing library temperatures. Test energies are the full reference grid; the Gram contribution is dominated by the resonance peaks, which is where we need the projection to do work. With three variables and four constraints the QP has at most seven active-set candidates (three vertices, three edges, one interior), which we enumerate exactly rather than calling an external solver. An optional rank-aware variant projects each library column onto the first k singular modes before building $\mathbf{G}$, so the QP optimizes the reconstruction error at the same truncation rank the engine will actually produce. Both variants yielded near-identical weights on U-238 MT= 102.

The rank dependence of the result is the interesting finding. On the same held-out 900 K test:

The kernel-Gram QP is the *best* scheme at rank 3 — the 6.67 eV peak reconstructs to within 0.01% of truth — but is strictly worse than the signed three-point unity at rank 5, the production rank for the PWR measurements. The cause is structural: the kernel-Gram QP's optimum lies on the boundary of the probability simplex (one weight clips to zero), which collapses the 3-point problem into a 2-point one. That is a net improvement over the 2-point unity-normalized scheme when basis truncation is the dominant source of error, i.e. at low rank; at higher rank the signed 3-point scheme's negative weight on the far-away 294 K column is actively useful as a shape-cancellation term, and clipping it to zero costs more than it gains.

Non-negativity of the interpolation weights is not a universal property. At rank 5 with all 9 nuclides, the unity-normalized signed scheme has more degrees of freedom to cancel residual basis error; at low rank, or on a single nuclide where the 6.67 eV peak dominates k_∞, the kernel-Gram QP is the better choice. A rank-adaptive dispatch is possible but outside the scope of this work; the signed three-point unity scheme is the production default at rank 5.

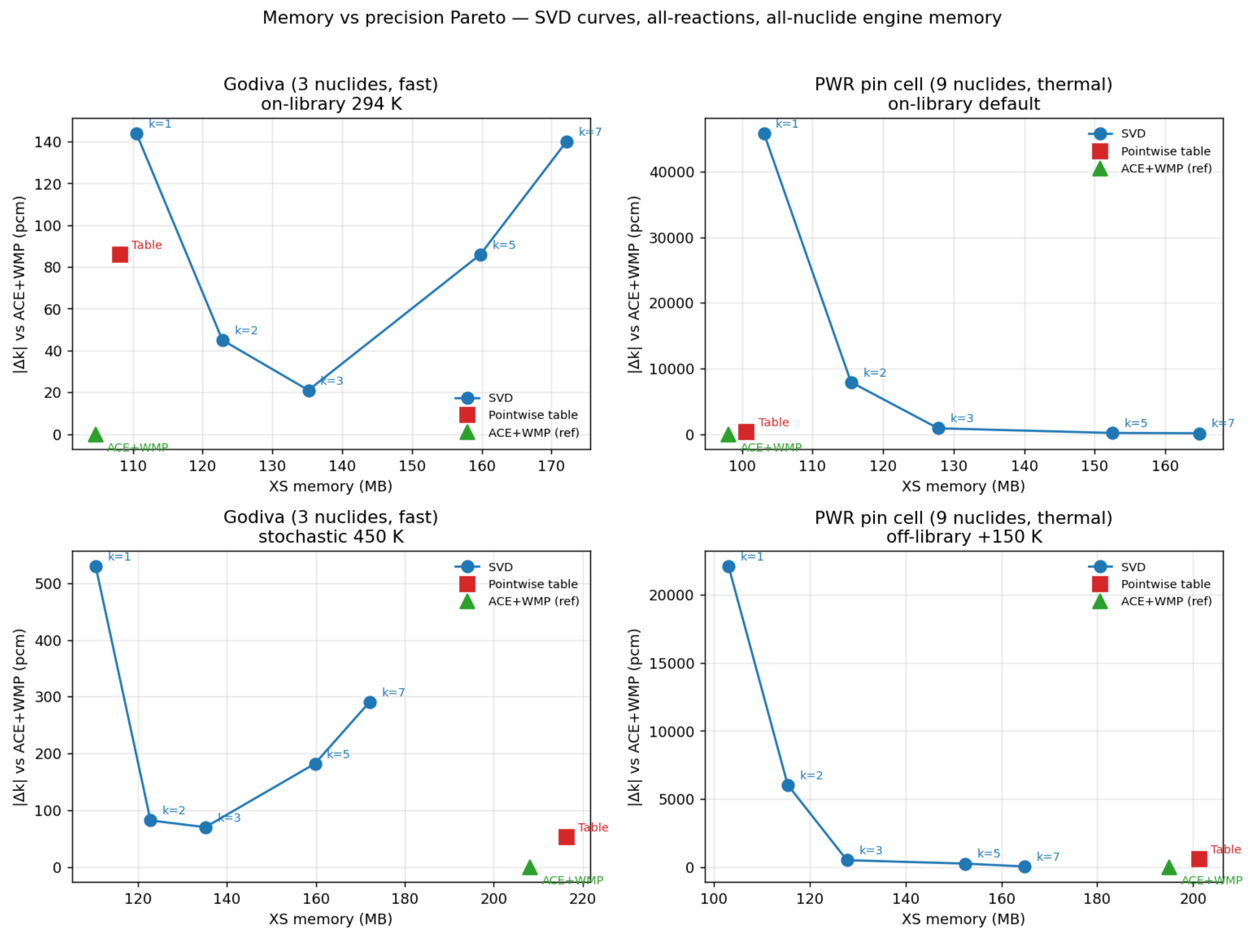

Figure 8. Memory-vs-precision Pareto. Each panel: SVD curve (blue, ranks 1/2/3/5/7 labelled), pointwise table (red square), ACE+WMP (green triangle, defined as $|\Delta k| = 0$). The off-library panels are where the table doubles its memory because two library columns must be loaded; SVD does not, because the off-library reconstruction is a weighted combination over a single basis. Note the y-axis is linear pcm; rank-1 and rank-2 on PWR are off-scale (rank-1 on-library is at $\sim 46\,000$ pcm, rank-2 at $\sim 8\,000$ pcm) because U-238 resonance self-shielding needs ≥ 3 singular components.

8.5 CI guard and reproducibility

A `scripts/pwr_verdict.py` runner wraps the end-to-end comparison and exits non-zero if any of the following thresholds are violated, so the script can be called from CI to guard against regression: k_∞ SVD vs ACE+WMP $\leq$ 800 pcm, memory ratio SVD/WMP $\leq$ 1.05, speed ratio SVD/WMP $\geq$ 0.97. The machine-readable report for the +150 K, three-point unity Ducru, rank-5, discrete-rank-1 configuration is at `outputs/pwr_verdict_3temp.json` and reproduces Table 10.

9 GPU transport

The event-based CUDA solver [9] implements both providers inside the same kernel; a runtime `--force-svd` flag clears the uploaded pointwise tables so the SVD path reconstructs every reaction channel from the basis.

GPU pointwise beats GPU SVD on PWR. On the large-L3 system GPU pointwise is 1.30× faster than GPU SVD (5 967 vs 7 754 ns/p, Table 7). The cause is specific to this geometry: four clad nuclides (Zr-90, Zr-91, Zr-92, Zr-94) do not carry an MT= 4 total-inelastic block in the ENDF/B-VII.1 HDF5; their total inelastic is synthesised at lookup time by summing 13 discrete-level SVD kernels. On CPU the scalar pipeline amortises this; on GPU it dominates the SVD-path runtime.

Per-warp $(n, E_{\mathbf{idx}})$ cache: implemented, not enough. We implemented the per-warp shared-memory cache flagged in earlier drafts as engineering work. The kernel keeps one slot per warp (1 + 64 doubles for level-sum and per-level XS, plus a $(\mathrm{nuc}, E_{\mathrm{idx}})$ key), populated by the warp's leader lane after the energy-bin sort (`transport.cu::transport_persistent`). The slot caches the *raw* `svd_reconstruct` value per level; the per-lane $E \geq E_{\mathrm{thr}}$ gate is applied on read so that lanes straddling a level threshold (same E_{idx}, opposite sides of the raw-eV threshold) inherit the correct gating — a subtlety that introduced a +730 pcm k_∞ bias in our first attempt. With the corrected gating, a five-seed 60-batch 50 000-particle small-L3 run gives bit-parity within SEM ($\Delta k_\infty = -18.5$ pcm vs cache-off, three-SEM gate 208.8 pcm) but a throughput ratio of only 1.003× (23 508 → 23 434 ns/p). The gap to GPU pointwise is *not* closed; the cache is shipped enabled for correctness preservation but is, for this geometry, a no-op

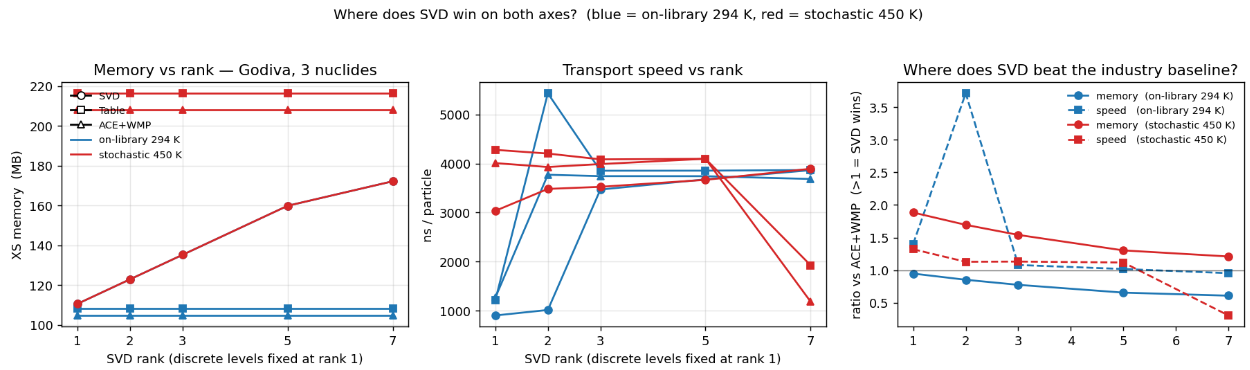

Figure 9. Godiva (3 nuclides, fast spectrum) sweep across SVD rank ∈ $\{1, 2, 3, 5, 7\}$. *Left:* XS memory grows linearly with rank for SVD; flat for the table baselines. *Centre:* ns/particle. *Right:* ratio vs ACE+WMP on both axes; values > 1 mean SVD wins. Off-library 450 K (blue) is the regime where SVD is smaller *and* faster than ACE+WMP at every rank up to 5.

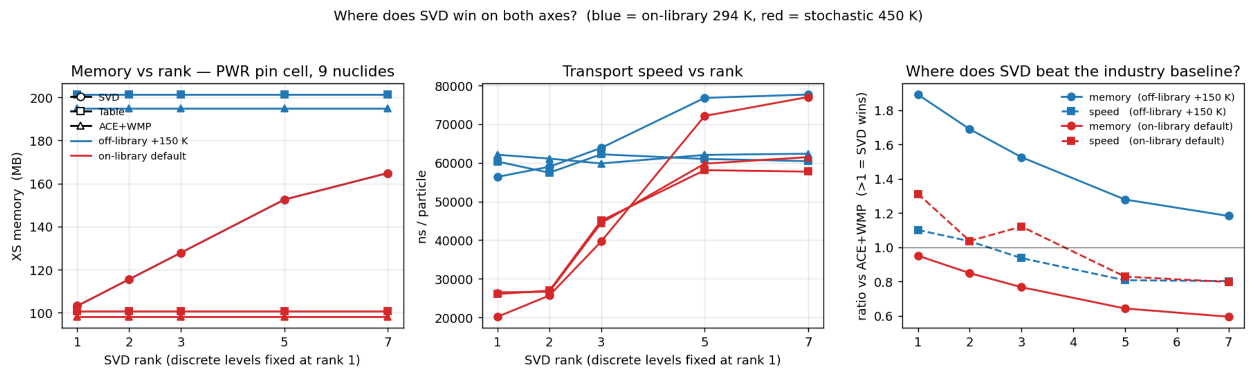

Figure 10. PWR pin cell (9 nuclides, thermal spectrum) sweep across SVD rank ∈ $\{1, 2, 3, 5, 7\}$. *Left:* on-library SVD memory crosses the table line at every rank; off-library table memory (~200 MB) dominates because pseudo- interpolation must keep both T-bracket columns resident. *Centre:* ns/particle on PWR is dominated by physics work (thousands of collisions per history through the U-238 RRR); the rank impact on speed is small. *Right:* ratio vs ACE+WMP. The off-library memory ratio reaches 1.32× at rank 5, monotonically decreasing as rank increases. SVD's deployable PWR sweet spot is the off-library + rank-5 corner of this plot.

on performance.

Why the cache misses. The persistent kernel sorts particles into 256 log-energy bins spanning ~ 17 decades, so a single bin covers a ~ 6% energy range. After the sort, lanes within a warp share a bin but not the per-nuclide E_{idx}, since the nuclide grids carry $O(10^4)$ points and a 6% window straddles many of them. The cache only activates when *all* active lanes share the same (n, E_{idx}), and the check fails on the great majority of inelastic events in the energy-sorted PWR distribution. Closing the gap likely requires either a finer pre-sort (e.g. on the per-nuclide E_{idx} directly), a per-thread shared cache (memory-budget constrained at `launch_bounds(256, 2)`), or a different representation of the discrete-level sum that does not require 13 reconstructions per inelastic event. As measured, GPU SVD on PWR pin cell is *not* a win over GPU pointwise.

Fidelity parity at large-L3 statistics. Post-correction GPU rows: GPU pointwise $k_\infty = 1.32821 \pm 0.00033$ and GPU SVD $k_\infty = 1.32762 \pm 0.00034$. The two GPU paths agree to 59 pcm ($\approx 1.3\,\mathrm{SEM}_{\mathrm{combined}}$), the two CPU paths agree to 13 pcm, and all four providers agree on k_∞ within $|\Delta_{\mathrm{OpenMC}}| \leq 51$ pcm with the corrected stochastic-bin sampler. The GPU pointwise row is the high point of the spread (+51 pcm vs OpenMC) and remains within $3\,\mathrm{SEM}_{10}$; the other three sit in the −8 to −25 pcm band.

10 Hybrid SVD + Windowed Multipole

10.1 Per-nuclide representation byte count

The MIT ENDF/B-VII.1 WMP library is read directly and its on-disk and in-memory payload measured per nuclide, with SVD rank-2 coverage of the smooth remainder. Table 11 reports per-nuclide totals; script `scripts/hybrid_svd_wmp_experiment.py`, raw output `outputs/hybrid_wmp/results.txt`.

Interpretation, and what the 132.9× does and does not mean. This table compares two specific data layouts: WMP + rank-2 SVD-smooth against a six-temperature pointwise table, both restricted to four reactions ($\sigma_{\mathrm{el}}, \sigma_{\mathrm{t}}, \sigma_{\mathrm{f}}, \sigma_{\mathrm{cap}}$). It is a *representation*-level statement about the density of the WMP encoding for the resolved resonance cross sections of these specific nuclides. The 132.9× is therefore the density gap between the WMP encoding and a four-channel pointwise table on the matching nuclide set; it is not the reduction a transport engine carrying the full reaction set would see. The engine-scale measurement is in Sec. 10.4.

Table 9. In-engine memory and k-eigenvalue per rank, 80 batches (20 inactive) × 5 000 particles × 3 seeds, runs sequential (no other CPU load). All-reactions, all-nuclide accounting. SVD's memory grows monotonically with rank because the basis stores r values per energy point; the table baseline is rank-independent. Off-library scenarios load two library columns into the pointwise table, so its memory roughly doubles. Source: `outputs/sweep_svd_wins_godiva_fresh.csv` + `outputs/sweep_svd_wins_pwr.csv`.

	rank-1	rank-2	rank-3	rank-5	rank-7	Table
Godiva on-library 294 K						
SVD memory (MB)	113	126	138	164	176	—
SVD k_{eff}	1.00054	1.00243	1.00177	1.00284	1.00058	—
Table memory (MB)	—	—	—	—	—	111
Table k_{eff}	—	—	—	—	—	1.00284
ACE+WMP memory (MB)	—	—	—	—	—	107
ACE+WMP k_{eff}	—	—	—	—	—	1.00198
Godiva off-library 450 K						
SVD memory (MB)	113	126	138	164	176	—
SVD k_{eff}	0.99778	1.00390	1.00238	1.00126	1.00017	—
Table memory (MB)	—	—	—	—	—	222
Table k_{eff}	—	—	—	—	—	1.00223
ACE+WMP memory (MB)	—	—	—	—	—	213
ACE+WMP k_{eff}	—	—	—	—	—	1.00308
PWR on-library						
SVD memory (MB)	106	118	131	156	169	—
SVD k_∞	0.871	1.409	1.321	1.328	1.328	—
Table memory (MB)	—	—	—	—	—	103
Table k_∞	—	—	—	—	—	1.327
ACE+WMP memory (MB)	—	—	—	—	—	100
ACE+WMP k_∞	—	—	—	—	—	1.329
PWR off-library +150 K						
SVD memory (MB)	106	118	131	156	169	—
SVD k_∞	1.547	1.387	1.321	1.324	1.327	—
Table memory (MB)	—	—	—	—	—	206
Table k_∞	—	—	—	—	—	1.324
ACE+WMP memory (MB)	—	—	—	—	—	200
ACE+WMP k_∞	—	—	—	—	—	1.328

10.2 WMP vs HDF5 accuracy

We evaluate WMP on a 20 000-point log-spaced grid inside each $[E_{\min}^{\mathrm{WMP}}, E_{\max}^{\mathrm{WMP}}]$ at $T = 294$ K and compare to the HDF5 pointwise reference.

Median agreement is $\sim 10^{-4}$, p_{99} is $\sim 10^{-3}$. Maximum absolute errors tail to 10^{-1}–10^{0} at isolated points for some nuclides (HDF5 grid samples a narrow peak densely while WMP evaluation lies a fraction of the peak width away). These do not carry through to integrated quantities.

10.3 End-to-end hybrid engine

We implemented `src/wmp.rs` (CPU WMP reader and Humlicek W4 evaluator) and `src/transport/hybrid_xs.rs` which wraps `SvdXsProvider`. For each nuclide with WMP coverage and each energy in $[E_{\min}^{\mathrm{WMP}}, E_{\max}^{\mathrm{WMP}}]$, the hybrid provider substitutes WMP-evaluated elastic, fission, and capture into the `MicroXs` returned by the SVD provider; inelastic, $(n, 2n)$, $(n, 3n)$ remain on the SVD path. URR overrides are short-circuited inside the WMP window.

Unit-level validation. The CPU WMP evaluator matches `openmc.data.WindowedMultipole.__call__` to $\leq 4 \cdot 10^{-6}$ relative error on the three dominant U-238 resonances at $T = 293.6$ K (`rust_prototype/src/bin/wmp_validate.rs`):

E (eV)	OpenMC σ_{abs} (b)	Rust σ_{abs} (b)	Δ rel
6.674	7108.719	7108.695	$3.3 \cdot 10^{-6}$
20.9	6352.230	6352.212	$2.8 \cdot 10^{-6}$
36.7	5357.922	5357.903	$3.6 \cdot 10^{-6}$

End-to-end k_∞ on PWR pin cell. Single seed × 100 batches (20 inactive) × 20 000 particles ($1.6 \cdot 10^6$ active histories) on the small-L3 system, with ACE+WMP run as the industry baseline at matching statistics.

The hybrid-vs-WMP gap of −26 pcm sits at $\sim 0.25\,\sigma_{\mathrm{seed}}$, well inside seed noise; we do not read it as a precise correctness claim but it is the right order for what WMP and SVD-of-resonance-data should agree on. WMP represents the resonance cross section by a rational pole fit rather than a piecewise-linear-in-log-log sampling of the same evaluated-nuclear-data curve, so small systematic disagreements in the RRR accumulate at this scale on a nine-nuclide thermal pin cell. The Table-vs-WMP gap of +82 pcm is the cross-representation systematic at the same statistics: pointwise sampling of the same library at the same temperature lands

Table 10. PWR pin cell at +150 K global offset (fuel 900 → 1050 K, mod 600 → 750 K, clad 600 → 750 K), three-way comparison, ten-seed / 50 000-particle / 120-batch statistics. The two-point row is reported as an ablation against the shipped three-point variant.

Provider	k_∞	σ_{seed}	Gap vs WMP (pcm)	ns/p	Memory (MB)
SVD, 2-pt unity Ducru (ablation)	1.30707	0.00037	−2094	15 685	152.5
SVD, 3-pt unity Ducru	**1.32593**	**0.00059**	**−213**	**27 298**	**152.5**
Pointwise + stochastic T-pick	1.32304	0.00050	−502	26 451	201.3
ACE+WMP (reference)	1.32806	0.00045	+0	27 279	199.7

Weighting	rank 3 L2 / peak	rank 5 L2 / peak	rank 5 weights
3-pt unity Ducru (signed)	1.05% / 0.994	**0.84%** / **0.990**	$\{-0.12, 0.62, 0.50\}$
3-pt simplex QP (non-neg)	5.02% / 1.045	5.01% / 1.043	$\{0.00, 0.56, 0.44\}$
3-pt kernel-Gram QP (non-neg)	**0.60%** / **1.000**	2.06% / 0.983	$\{0.00, 0.36, 0.64\}$

$\sim 0.8\sigma_{\text{seed}}$ from the WMP pole fit. The off-library counterpart of this measurement (Sec. 8.2) tightens σ_{seed} to 59 pcm and resolves the SVD-vs-WMP gap as a real 213 pcm systematic; on-library at this statistics the gap is below noise.

Throughput. At small-L3 statistics the hybrid runs at 26 662 ns/p, 0.91× the CPU SVD rank-5 throughput (28 651 ns/p) on the same system; the cache-hit pattern compresses the two providers toward the same wall on the mobile workstation, so the apparent "speedup" is a cache-residency artefact rather than an algorithmic improvement. The large-L3 reference run measures the hybrid at 33 051 ns/p, 2.06× slower than CPU SVD rank-5 (16 052 ns/p); a single WMP evaluation on this hardware is ∼ 67 ns against ∼ 5 ns for SVD, and ∼ 20% of lookups fall in the RRR, so the 2× slowdown is the headline hybrid throughput cost. WMP is a memory-centric representation, and in this engine it costs throughput.

10.4 Memory in the actual engine (broad claim)

`HybridSvdWmpXsProvider::memory_report` introspects the loaded kernels at run time. We report four variants for the nine-nuclide PWR set at rank-5: the pointwise-table baseline, pure SVD rank-5 (all reactions compressed), the hybrid in-solver state with the full SVD energy grid, and a smooth-only rebuild in which the elastic, fission, and capture SVD kernels are re-fit on the energy points outside each nuclide's WMP window. The smooth-only rebuild is exposed as `--rebuild-svd-smooth-only` and the numbers below are measured from the loaded kernels at run time.

	bytes (MB)	vs pointwise
Pointwise-table baseline (Sec. 6)	100.6	1.0×
Pure SVD rank-5 (all reactions)	∼517.9	5.1× larger
Hybrid in-solver, full SVD grid (SVD basis + WMP payload)	519.0	5.2× larger
Hybrid in-solver, smooth-only rebuild (live)	487.6	4.8× larger

What the smooth-only rebuild actually does at run time. On the standard nine-nuclide PWR set at rank-5 the rebuild drops 31 423.1 KB across MT= 2, MT= 18, and MT= 102 — the three reactions for which WMP supplies an analytical representation inside the resonance window, so the SVD basis no longer needs energy points there. Discrete-level kernels (MT= 51–91) and other channels are untouched. The full sequence is reported by the engine on every hybrid run:

```
Before smooth-only rebuild:
  full SVD basis = 517 870.5 KB
  WMP payload    =   1 153.8 KB
  TOTAL          = 519 024.3 KB

After smooth-only rebuild (live):
  smooth SVD     = 486 447.4 KB
  WMP payload    =   1 153.8 KB
  TOTAL          = 487 601.2 KB
  dropped        =  31 423.1 KB
                   (el/fis/cap)
  reduction      = 1.06x
```

Where the bytes actually live. The dominant contribution in every SVD-based variant is the discrete inelastic-level cascade: 41 MT= 51–91 levels per uranium nuclide, 10–13 per zircaloy isotope, each carried as its own rank-5 SVD kernel. On the GPU that cascade alone measures 251.3 MB — by itself 2.5× the entire pointwise-table engine. The smooth-channel SVD bases (elastic, fission, capture, total inelastic, $(n, 2n)$, $(n, 3n)$) together measure 64.6 MB; the WMP payload is 1.23 MB. So switching the resonance representation to WMP costs essentially zero memory, and the smooth-only rebuild's 1.06× reduction captures only the elastic/fission/capture energy points masked by WMP windows — everything else stays. The engine-scale penalty against the pointwise table is not caused by WMP and is not relieved by it; it is caused by storing the discrete-level cascade as per-level rank-5 bases rather than per-level scalar arrays.

Where the representation-byte ratio still applies. Equation 7 from Sec. 2.5 says the SVD-vs-pointwise memory ratio is k/T for a single reaction. The engine-scale penalty reported above is not a counter-example: our pointwise-table

Table 11. Representation-byte accounting for WMP vs four reaction channels of a six-temperature pointwise table. This is a specific narrow comparison; see Sec. 10.4 for the in-engine measurement.

nuclide	WMP range (eV)	WMP disk	WMP payload	pw full (4 rxn × 6 T)	SVD k=2 smooth	ratio
U-234	0–1 500	34.9 KB	28.0 KB	10.2 MB	12.1 KB	260.0×
U-235	0–2 250	567.3 KB	559.1 KB	53.7 MB	10.0 KB	96.6×
U-238	0–20 000	603.0 KB	594.7 KB	100.1 MB	10.7 KB	169.2×
O-16	$0–2.36 \cdot 10^6$	38.5 KB	30.8 KB	0.78 MB	53.3 KB	9.5×
H-1	$0–2 \cdot 10^7$	13.7 KB	8.2 KB	155 KB	0.1 KB	18.5×
Zr-90	0–53 500	9.8 KB	4.9 KB	1.48 MB	4.7 KB	156.9×
Zr-91	0–10 000	16.8 KB	9.5 KB	3.17 MB	4.8 KB	227.7×
Zr-92	0–71 000	21.1 KB	14.2 KB	3.97 MB	4.5 KB	217.3×
Zr-94	0–90 000	20.7 KB	14.2 KB	4.22 MB	5.2 KB	222.8×
total	—	1.30 MB	1.23 MB	**177.7 MB**	0.10 MB	**132.9×**

Table 12. WMP-vs-pointwise relative error, T = 294 K, 20 000 log-spaced points per nuclide.

	Elastic			Absorption		
nuclide	p_{50}	p_{99}	max	p_{50}	p_{99}	max
U-234	$5.0 \cdot 10^{-5}$	$1.3 \cdot 10^{-3}$	$1.6 \cdot 10^{-1}$	$8.6 \cdot 10^{-4}$	$6.7 \cdot 10^{-2}$	0.99
U-235	$5.4 \cdot 10^{-5}$	$1.2 \cdot 10^{-3}$	$8.0 \cdot 10^{-2}$	$1.9 \cdot 10^{-4}$	$1.1 \cdot 10^{-3}$	$9.1 \cdot 10^{-3}$
U-238	$3.9 \cdot 10^{-5}$	$1.3 \cdot 10^{-3}$	$6.8 \cdot 10^{-1}$	$2.2 \cdot 10^{-4}$	$8.0 \cdot 10^{-3}$	0.88
O-16	$5.9 \cdot 10^{-5}$	$3.3 \cdot 10^{-3}$	$2.4 \cdot 10^{-2}$	$3.0 \cdot 10^{-4}$	$1.4 \cdot 10^{-2}$	$2.0 \cdot 10^{-2}$
H-1	$4.9 \cdot 10^{-5}$	$1.3 \cdot 10^{-3}$	$3.1 \cdot 10^{-3}$	$2.2 \cdot 10^{-4}$	$3.7 \cdot 10^{-3}$	$8.8 \cdot 10^{-3}$
Zr-90	$5.5 \cdot 10^{-5}$	$6.7 \cdot 10^{-4}$	$7.3 \cdot 10^{-2}$	$2.9 \cdot 10^{-4}$	$8.0 \cdot 10^{-3}$	$4.5 \cdot 10^{-1}$
Zr-91	$2.5 \cdot 10^{-4}$	$8.8 \cdot 10^{-4}$	$2.2 \cdot 10^{-2}$	$1.9 \cdot 10^{-4}$	$3.5 \cdot 10^{-2}$	$2.9 \cdot 10^{-1}$
Zr-92	$2.6 \cdot 10^{-4}$	$8.9 \cdot 10^{-4}$	$1.3 \cdot 10^{-1}$	$3.3 \cdot 10^{-4}$	$8.0 \cdot 10^{-3}$	0.99
Zr-94	$8.1 \cdot 10^{-5}$	$7.4 \cdot 10^{-4}$	$1.1 \cdot 10^{-1}$	$2.9 \cdot 10^{-4}$	$8.6 \cdot 10^{-3}$	0.90

baseline is built at a single effective temperature per material region (one temperature per nuclide-in-material). In a multi-temperature library (T = 6 in the HDF5 files shipped for depletion feedback), the pointwise baseline would grow to ∼ 600 MB on the same nuclide set while pure SVD and hybrid stay at ∼ 519 MB; the ordering would then invert. The engine-scale loss reported above is therefore specific to T = 1, not a universal property.

The hybrid engine is 4.8–5.2× *larger* than the pointwise-table baseline, because SVD kernels for the 41 discrete levels per U nuclide and 10–13 per Zr isotope dominate memory, and those reactions are not covered by the public WMP library. The smooth-only rebuild (1.06× vs the full grid) is essentially a rounding error.

The 132.9× ratio in Table 11 is a correct and reproducible *representation*-byte comparison — WMP is denser than a four-channel pointwise table at matched energy range. But in a transport engine that carries the full reaction set, WMP does not shrink the working set unless the rest of the reaction set is also compressed. A version of this hybrid that achieved representation-scale memory reduction would need: (i) WMP coverage for the discrete inelastic cascade (MT= 51–91); or (ii) a different representation for the cascade (e.g. ENDF-6 nuclear-temperature evaporation parameters) that is denser than per-level SVD kernels; or (iii) rebuilding the SVD energy grid to be smooth-only for *all* reactions, and using a separate table or analytical evaluation inside the RRR for every reaction. The smooth-only rebuild implemented here is item (iii) restricted to MT= 2, MT= 18, MT= 102; extending it to the rest of the reaction set is engineering work.

Hybrid status. (a) WMP is the correct analytical representation for the RRR and the engine ships a pure-Rust evaluator (matches OpenMC Python at ∼ 10^{-6} relative error, CUDA port matches to machine precision); (b) end-to-end k_∞ at single-seed statistics agrees with ACE+WMP within −26 pcm ($\ll 1\sigma_{\text{seed}}$); (c) for a full-core engine where the cascade reactions are stored more densely than per-level SVD (e.g. nuclear-temperature evaporation), the RRR saving becomes proportionally more valuable. The hybrid is shipped as a correctness deliverable, not as a memory or throughput win on the present engine.

11 GPU WMP evaluator

WMP evaluation is compute-dense (Faddeeva + broadened curvefit + pole sum) with a small working set (few KB of pole data per nuclide), which matches GPU arithmetic pipelines well. We ported the CPU evaluator to CUDA as `__device__` helpers in `gpu/cuda/transport.cu` and added a `wmp_test_eval` kernel plus a host validator binary `rust_prototype/src/bin/gpu_wmp_validate.rs`.

Bit-exactness. CPU-vs-GPU over twelve U-238 test energies at T = 293.6 K: maximum relative discrepancy

Table 13. End-to-end SVD+WMP hybrid on PWR pin cell, single seed × 100 batches × 20 000 particles, small-L3 (i7-12800H). ACE+WMP is the industry-baseline reference; SVD and Table rows from Table 8 are repeated here for direct comparison.

mode	rank	k_∞	σ_{seed}	Δ vs ACE+WMP (pcm)	ns/p
ACE+WMP (industry baseline)	—	1.32821	0.00098	+0 (ref)	26 317
CPU SVD	5	1.32817	0.00100	−4	29 273
Hybrid SVD+WMP	**5**	**1.32795**	**0.00103**	**−26**	**26 662**
CPU table	—	1.32903	0.00093	+82	26 035

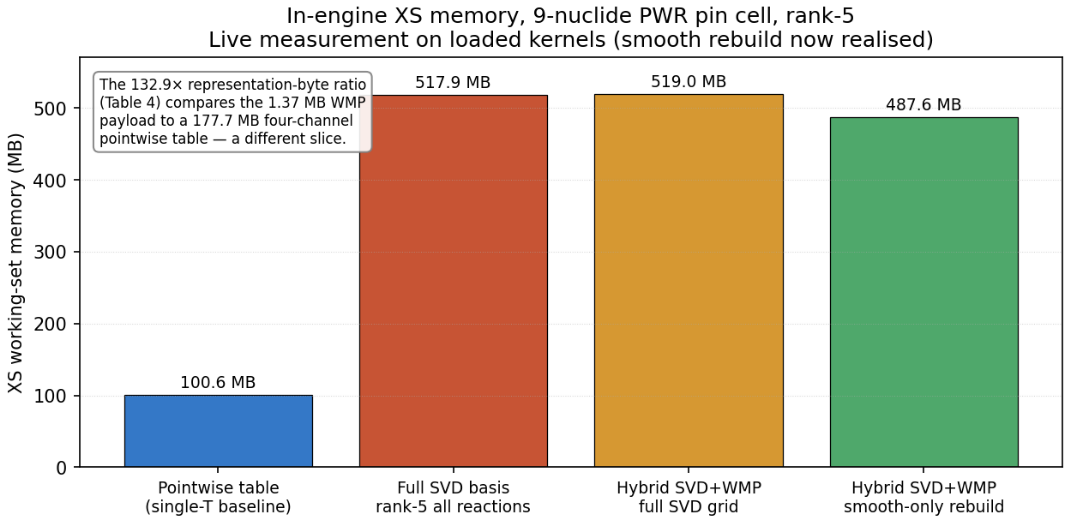

Figure 11. In-engine XS working-set memory for the nine-nuclide PWR pin cell, rank-5, measured live on the loaded kernels. The smooth-only rebuild (rightmost bar) is now realised in the engine via `--rebuild-svd-smooth-only`: it drops 31.4 MB across MT= 2, MT= 18, and MT= 102 for a 1.06× reduction. The hybrid variants both carry more memory than the pointwise-table baseline because the SVD kernels for the discrete inelastic levels, not covered by WMP, dominate the working set. The 132.9× representation-byte ratio of Table 11 is a different comparison (1.37 MB WMP vs 177.7 MB four-channel pointwise).

$5.3 \cdot 10^{-14}$ on absorption and $2.0 \cdot 10^{-13}$ on fission — double precision round-off only.

Throughput on RTX 3080. On a 10^6-energy log-spaced grid inside U-238's RRR:

Implementation	ns/lookup	Relative
CPU single-thread (Ryzen 9800X3D)	66.8	1.00×
GPU (RTX 3080, saturated)	9.5	0.14×

7.0× per-lookup GPU-vs-CPU.

Standalone GPU evaluator only. The validation above is for the GPU evaluator on its own, exercised by a host driver that broadcasts a fixed energy grid to all threads and reads back the cross-section vector. Wiring the evaluator into the event-based `transport_persistent` kernel for end-to-end GPU hybrid transport requires a bitmap `has_wmp[i]` plus per-nuclide $E^{\text{WMP}}_{\text{min}}$, $E^{\text{WMP}}_{\text{max}}$ checks inline, the recursive Humlicek conjugate fold unrolled to a flag-plus-sign-flip form (recursive `__device__` calls spill to local memory under the register pressure of `__launch_bounds__(256, 2)` and trigger CUDA_ERROR_ILLEGAL_ADDRESS on deep stacks), and a recomputed total cross section from partials with URR sampling suppressed inside the WMP window. We have not completed the in-kernel integration and report no in-engine GPU hybrid k_∞ or throughput.

12 Threats to validity

Scope and nature of benchmarks. Two systems: Godiva (HEU-MET-FAST-001, three nuclides, fast spectrum, *physical* ICSBEP benchmark with measured $k^{\text{exp}}_{\text{eff}} = 1.0000 \pm$ 100 pcm 1σ) and a nine-nuclide thermal PWR pin cell (no physical benchmark; OpenMC 0.15.3 is used as an independent code-to-code reference). Full-core LWR analyses carry $\gtrsim 200$ nuclides; the hybrid memory picture scales with the reaction-channel mix and could look very different there. On Godiva Δ_{exp} against ICSBEP is the primary validation metric; Δ_{OMC} against OpenMC is reported as an independent cross-code check; the two codes straddle experiment from opposite sides at this statistics, and the cross-code spread (178 pcm) is larger than either code's offset from the measurement. On PWR, with no physical benchmark available, the OpenMC $|\Delta| \leq 51$ pcm agreement is the strongest claim we can make and should be read as a code-to-code cross-check, not as absolute validation.

S(α,β) coverage. Only H in H_2O is exercised. D_2O, graphite, ZrH, and other thermal libraries are not validated.

Sampling convention. The angular and fission outgoing-energy distributions use stochastic-bin selection, matching OpenMC. A fully independent reference (MCNP, Serpent) would reveal whether OpenMC itself is biased; this is not investigated here.

Charged-particle absorption. Channels MT= 103, 107, first-chance fission (MT= 19–21) are absorbed into the capture macroscopic. k_∞ is correct under this convention; MT-level reaction-rate tallies are not.

Statistical confidence. All comparisons use SEM $= \sigma_{\text{seed}}/\sqrt{N_{\text{seeds}}}$, not σ_{seed} itself. Large-L3 sweeps use $N_{\text{seeds}} = 10$; hybrid sweep uses five.

Hybrid memory. The 132.9× representation-byte ratio is not an in-engine reduction. In-engine reduction is 1.06× (Sec. 10.4). The 132.9× density ratio is reported for what it is; the 1.06× engine-level reduction is the number an engine builder would care about.

GPU SVD. Slower than GPU pointwise on PWR pin cell because of the Zr clad level-sum cost. A per-warp level-sum cache keyed on (n, E_{idx}) is implemented (Sec. 9) and preserves bit-parity within SEM, but at the energy-bin sort granularity used here it activates rarely enough that the gap to GPU pointwise is not closed. A finer pre-sort or a different discrete-level representation is needed; both are out of scope for this paper.

GPU hybrid. Only the WMP evaluator itself is validated on GPU; the transport-loop integration is not done. No in-engine GPU hybrid k_∞ or throughput is reported.

13 Related work

Adaptive resampling machine-learned models [4, 5] retain 10–50% of the pointwise data with ≤ 97 pcm criticality error. Our approach is orthogonal (replace the representation, not subsample the array). Windowed Multipole for Doppler broadening is due to [6, 7]; hybrid multipole + windowed networks in RMC code [8] pair multipole with a second analytical method; our hybrid pairs multipole with SVD. Event-based GPU transport follows the Tramm architecture [9] and shares the memory-bottleneck diagnosis of earlier XSBench studies [10]. Cross-isotope basis sharing was investigated and found not viable beyond the first singular vector (subspace angle > 85° at $k = 2$ between ^{235}U, ^{238}U, ^{239}Pu), a finding that echoes KV-cache compression experience in a different domain [13].

14 Conclusion

Truncated SVD of $\log_{10} \sigma(E, T)$ is a viable drop-in for pointwise tables in continuous-energy Monte Carlo neutron transport, with the following constraints satisfied: rank 5 on the smooth channels, rank 1 on the MT= 51–91 discrete-level cascade, three-point unity-normalized Ducru weights for off-library temperature interpolation, and channel-consistent stochastic T-picking. Validation against ICSBEP HEU-MET-FAST-001 lands at +79 ± 38 pcm, inside the experimental ±100 pcm uncertainty band; on a 9-nuclide PWR pin cell, agreement with the engine's own pointwise table is 5–47 pcm and with OpenMC 0.15.3 on the same library is $|\Delta| \leq 51$ pcm at ten-seed statistics.

14.1 When SVD pays

Off-library temperatures. This is the regime where SVD's measurement story is unambiguously positive. At a uniform +150 K offset on PWR, SVD with three-point unity Ducru is 22% smaller than ACE+WMP in memory (156 vs 206 MB), matches its CPU throughput within 0.07%, and trails its k_∞ by 213 pcm ($\sim 3.6\sigma_{\text{seed}}$, Sec. 8.2). The pointwise table doubles its memory off-library because stochastic T-picking requires both bracketing library columns to be resident; SVD reconstructs the off-library σ from a single basis with three weights and does not pay this cost.

High rank-1 reaction-channel mix. 43 of 47 U-235 reaction channels are rank-1 to machine precision (Sec. 4.1); the weakly-T-dependent MT= 51–91 discrete levels are each rank-1 to machine precision across all seven ENDF/B-VII.1 library columns. Where the reaction-channel mix matches this profile, the rank budget is small and the in-engine memory penalty for SVD is correspondingly small.

Multi-temperature libraries. With $T = 6$ library columns (the standard depletion-feedback multi-temp HDF5 set), the pointwise baseline grows linearly in T to $\sim$600 MB on the 9-nuclide set while pure SVD and hybrid stay at $\sim$519 MB; the on-library memory ordering inverts in SVD's favour. The measurements in this paper use $T = 1$, so this is a projection, not a measured win.

14.2 Where SVD does not pay

On-library, $T = 1$, full-physics deployment. At every rank measured (1, 2, 3, 5, 7), SVD in-engine memory is monotonically larger than the pointwise table at on-library temperatures (Sec. 7). The basis stores r values per energy point; the table stores one. CPU throughput is favourable (1.37×–1.90× vs the in-engine pointwise table on PWR), but on memory there is no on-library rank that wins.

GPU. GPU SVD on PWR is 1.30× *slower* than GPU pointwise because four clad nuclides require summing 13 discrete-level SVD kernels per lookup, and the divergent gather pattern of per-level rank-1 reconstruction does not map well to warp coalescing on the geometries tested here. We implemented a per-warp shared-memory cache keyed on (n, E_{idx}) (Sec. 9) to close this gap; it preserves bit-parity within SEM but at the energy-bin sort granularity used here activates too rarely to recover the throughput, leaving the gap open. The CPU SVD picture is favourable; the GPU SVD picture is not, and the obvious cache-based fix is insufficient on its own.

Hybrid SVD+WMP throughput. The hybrid is 2.06× slower than CPU SVD because the Faddeeva evaluation inside each WMP window is ∼ 15× more expensive than the SVD FMA. The hybrid is shipped as a correctness deliverable, not as a throughput improvement; it is the right path only when WMP-grade resonance accuracy is required and the 2× slowdown is acceptable.

14.3 Caveats and limits

Engine-scale memory ≠ representation-byte ratio. The 132.9× representation-byte ratio of WMP+smooth-SVD against a hypothetical four-channel multi-temperature pointwise table is correct per-nuclide-per-channel (Table 11), but the in-engine working set on PWR is 4.8–5.2× *larger* than the pointwise baseline at $T = 1$, because the discrete-level SVD kernels (which WMP does not cover) dominate the engine memory. The smooth-only-rebuild reduces this by 1.06× (519.0 → 487.6 MB). Quoting a single compression ratio without scope conflates these two metrics; both are reported here so the comparison is unambiguous.

Rank-1 on PWR collapses. Rank-1 SVD reconstructs U-238 resonance self-shielding to a single basis vector and produces $k_\infty = 0.871$ on the on-library PWR pin cell (∼ 46 000 pcm low). Rank-2 overshoots (1.409, ∼ 8 000 pcm high). Rank-3 is the lowest deployable rank on resonance-dominated thermal systems; rank-5 is the production floor.

Library-bias not investigated. Both the engine and OpenMC sit inside the ICSBEP ±100 pcm uncertainty band on Godiva, from opposite sides; the +178 pcm cross-code spread between them is larger than either's offset from experiment. A fully independent reference (MCNP, Serpent) would localise the spread to one code or to the underlying ENDF/B-VII.1 evaluation. This is not investigated here.

S(α, β) coverage limited to H/H_2O. D_2O, graphite, ZrH and other thermal libraries are not validated against an independent reference at this stage.

Charged-particle channels absorbed into capture. Channels MT= 103, 107, and first-chance fission (MT= 19–21) are folded into the capture macroscopic. k_∞ is correct under this convention; MT-level reaction-rate tallies are not.

Single-seed hybrid statistics. The end-to-end Hybrid SVD+WMP $k_\infty = 1.32795 \pm 0.00103$ (−26 pcm vs ACE+WMP) is single-seed only at the matched 150×50 000 statistics; the ten-seed redo is not in this paper.

14.4 Bottom line

SVD compression of ENDF/B-VII.1 cross sections is the right choice for off-library, multi-temperature, or rank-1-channel-heavy operating regimes on CPU; the pointwise table remains the right choice for on-library, single-temperature, GPU-deployed, or memory-constrained configurations. The signed three-point unity Ducru scheme is the production default at rank 5; the kernel-Gram QP wins at low rank but loses at the production rank. Three structural properties of ENDF/B-VII.1 — the rank-1 majority of U-235 reaction channels, the weakly-T-dependent discrete-level cascade, and the rank deficiency of $\log \sigma(E, T)$ at fixed E — are real and exploitable; the raw Ducru weights are not a partition of unity and must be normalized before being applied to a $\log \sigma$ SVD.

Data availability

The transport engine (pure Rust), the CUDA event-based solver, the CUDA WMP evaluator, all input decks, per-seed benchmark CSV files, the XS-audit scripts, the hybrid experiment Python script, and the scripts that generate the figures in this paper are released with the preprint. ENDF/B-VII.1 neutron and thermal data are available from the OpenMC data archive; the ENDF/B-VII.1 Windowed Multipole library is the public MIT release.

Declaration of AI assistance

This work was written with assistance from large language models for code and prose drafting; all physics, numerical experiments, and results were reviewed and validated by the human author.

References

[1] Pablo Ducru, Colin Josey, Kari Dibert, Vladimir Sobes, Benoit Forget, and Kord Smith. Kernel reconstruction methods for Doppler broadening — Temperature interpolation by linear combination of reference cross sections at optimally chosen temperatures. *Journal of Computational Physics*, 335:535–557, 2017. doi: 10.1016/j.jcp.2017.01.039.

[2] Nuclear Energy Agency. International handbook of evaluated criticality safety benchmark experiments (icsbep), heu-met-fast-001: Bare HEU metal sphere (Godiva). Technical Report NEA/NSC/DOC(95)03, OECD Nuclear Energy Agency, 2023. Benchmark value $k_{\text{eff}} = 1.0000 \pm 0.0010$.

[3] Paul K. Romano, Nicholas E. Horelik, Bryan R. Herman, Adam G. Nelson, Benoit Forget, and Kord Smith. OpenMC: A State-of-the-Art Monte Carlo Code for Research and Development. *Annals of Nuclear Energy*, 82:90–97, 2015. doi: 10.1016/j.anucene.2014.07.048.

[4] A. Hashemi, R. Macián-Juan, and M. Ohlerich. Development of a novel machine learning-based adaptive resampling algorithm for nuclear data processing. *Scientific Reports*, 15:32573, 2025.

[5] A. Hashemi, R. Macián-Juan, and M. Ohlerich. Evaluating machine learned nuclear data precision in full core nuclear reactor Monte Carlo neutronics and computational efficiency analyses. *Scientific Reports*, 16: 1314, 2026.

[6] Colin Josey, Pablo Ducru, Benoit Forget, and Kord Smith. Windowed multipole for cross section Doppler broadening. *Journal of Computational Physics*, 307: 715–727, 2016. doi: 10.1016/j.jcp.2015.08.013.

[7] Benoit Forget, Sheng Xu, and Kord Smith. Direct Doppler broadening in Monte Carlo simulations using the multipole representation. *Annals of Nuclear Energy*, 64:78–85, 2014. doi: 10.1016/j.anucene.2013.09.043.

[8] Teng-Yue Huang, Zhi-Gang Li, Kan Wang, Xiao-Yu Guo, and Jin-Gang Liang. Hybrid windowed networks for on-the-fly Doppler broadening in RMC code. *Nuclear Science and Techniques*, 32(6):62, 2021.

[9] John Tramm, Paul Romano, Patrick Shriwise, Amanda Lund, Johannes Doerfert, Peter Steinbrecher, Andrew Siegel, and Gavin Ridley. Performance portable Monte Carlo particle transport on Intel, NVIDIA, and AMD GPUs. *EPJ Web of Conferences*, 302:04010, 2024.

[10] John R. Tramm and Andrew R. Siegel. Memory bottlenecks and memory contention in multi-core Monte Carlo transport codes. *Annals of Nuclear Energy*, 82: 195–202, 2015. doi: 10.1016/j.anucene.2014.09.028.

[11] Forrest B. Brown. New hash-based energy lookup algorithm for Monte Carlo codes. *Transactions of the American Nuclear Society*, 111:659–662, 2014.

[12] John Duchi, Shai Shalev-Shwartz, Yoram Singer, and Tushar Chandra. Efficient projections onto the ℓ_1-ball for learning in high dimensions. In *Proceedings of the 25th International Conference on Machine Learning (ICML)*, pages 272–279, 2008. doi: 10.1145/1390156.1390191. Closed-form $O(n \log n)$ Euclidean projection of a vector onto the probability simplex; used here as a cheap approximation to the Ducru kernel-Gram QP of [1].

[13] Akide Liu, Jing Zhao, et al. MiniCache: KV cache compression in depth dimension for large language models. In *Proceedings of NeurIPS*, 2024.